# Finite Element Analysis of Low-Velocity Impact (LVI) Damage in Bio-inspired Helicoid CFRP Laminates for Thin- and Thick-Wall Structures

IMPERIAL

TexComp 9-11 September 2026 Bristol, UK

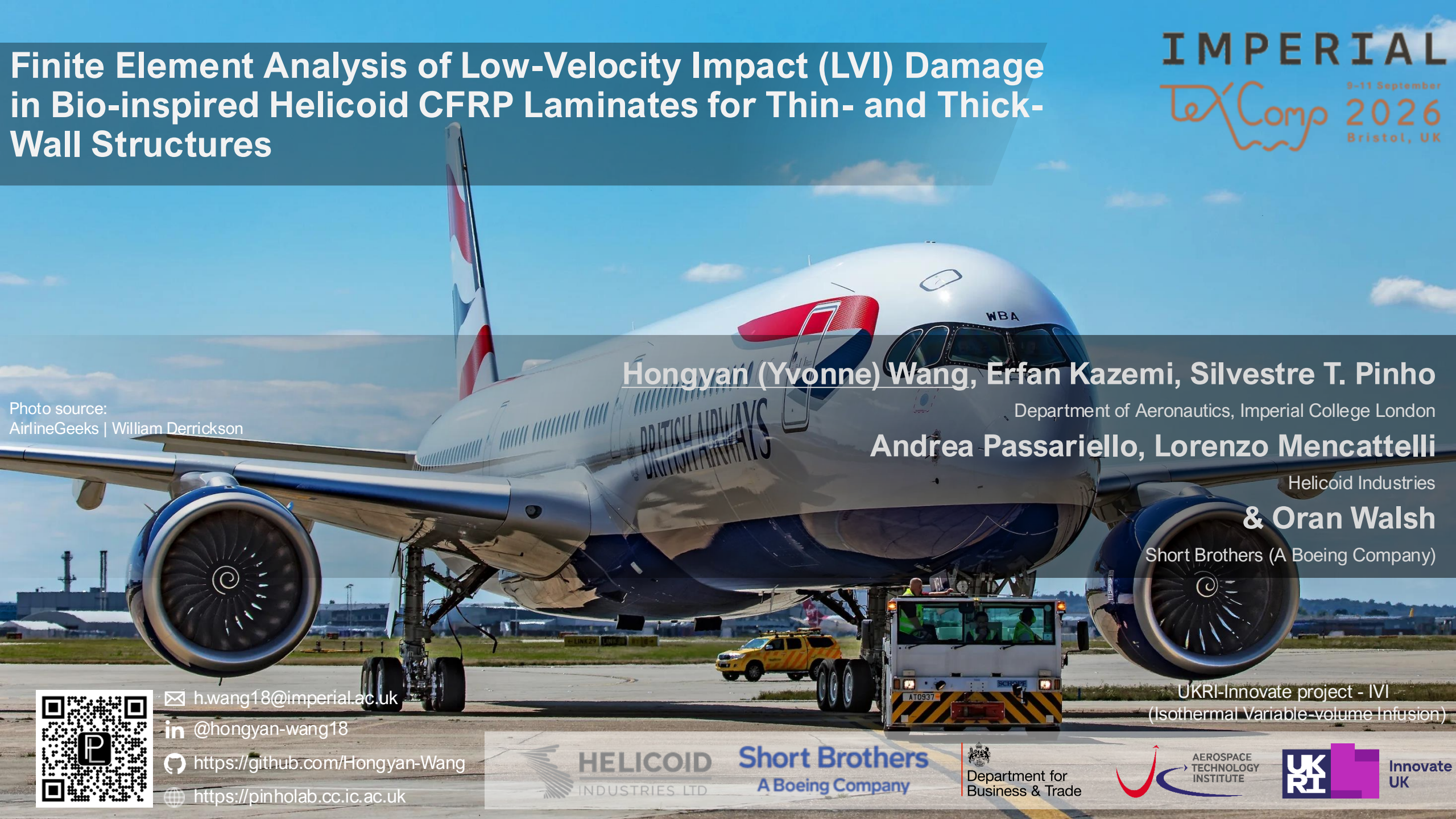



Photo source:
AirlineGeeks | William Derrickson

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Helicoid Industries

& Oran Walsh

Short Brothers (A Boeing Company)

UKRI-Innovate project - IVI
(Isothermal Variable-volume Infusion)

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HELICOID INDUSTRIES LTD

Short Brothers A Boeing Company

Department for Business & Trade

AEROSPACE TECHNOLOGY INSTITUTE

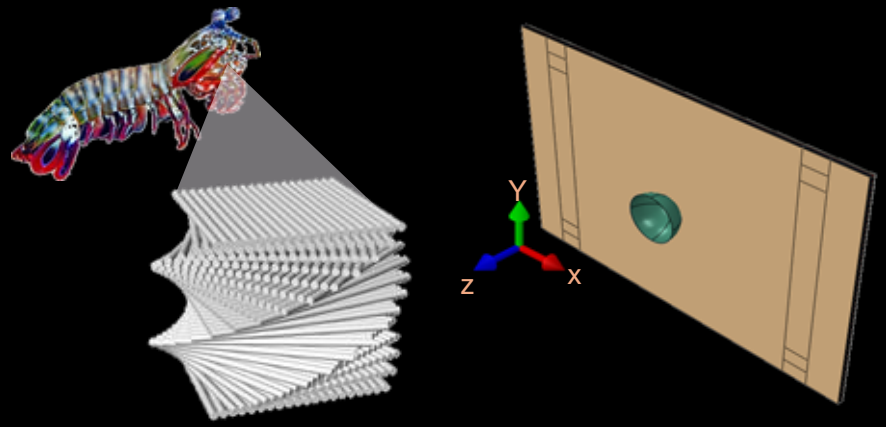
UKRI Innovate UK

# Outline

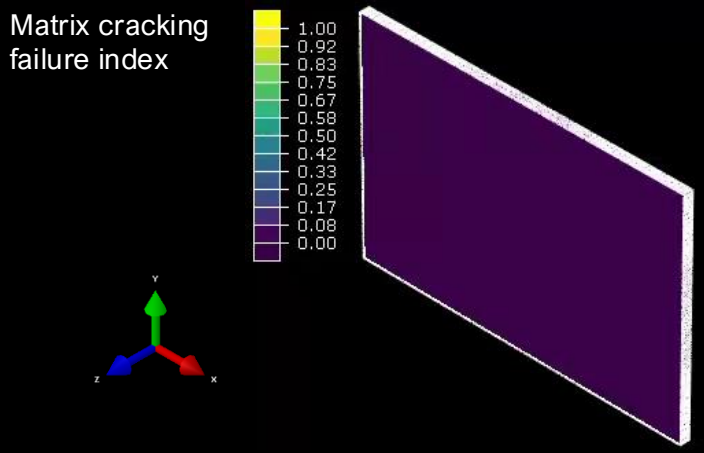
## Background & Motivation



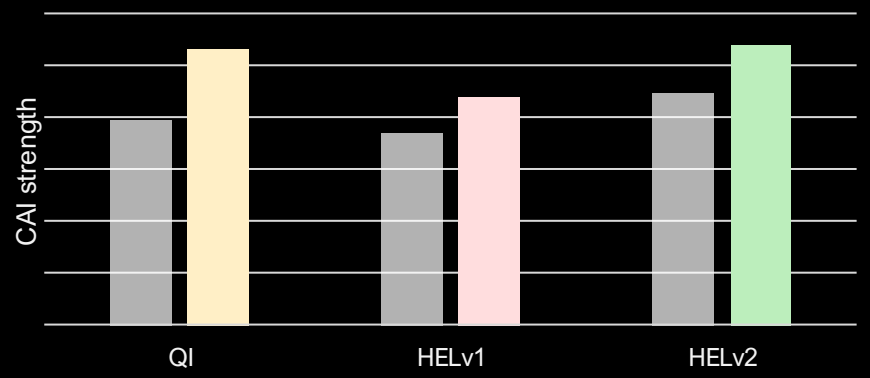

## Design & Methodology




## Results & Discussion




## Summary & Conclusions



# Background & Motivation




Photo source:
AirlineGeeks | William Derrickson

# Background & Motivation

**>50 wt%**

Carbon Fibre-Reinforced Polymer (CFRP) composites in A350 and B787

**20-22%**

Weight reduction achieved

**20-25%**

Fuel savings achieved [1]

## Current challenges

- Complex, multi-axial, and uncertain loading scenarios
- Exposure to harsh environments and impact threats
- Limited damage tolerance in conventional CFRP layups

## Proposed approach

Bio-inspired laminate design to enhance impact damage tolerance and maintain structural integrity under critical loading conditions.

Photo source: AirlineGeeks | William Derrickson

[1] Zhang, J., et al. (2023). "Past, present and future prospective of global carbon fibre composite developments and applications." Composites Part B: Engineering 250: 110463.



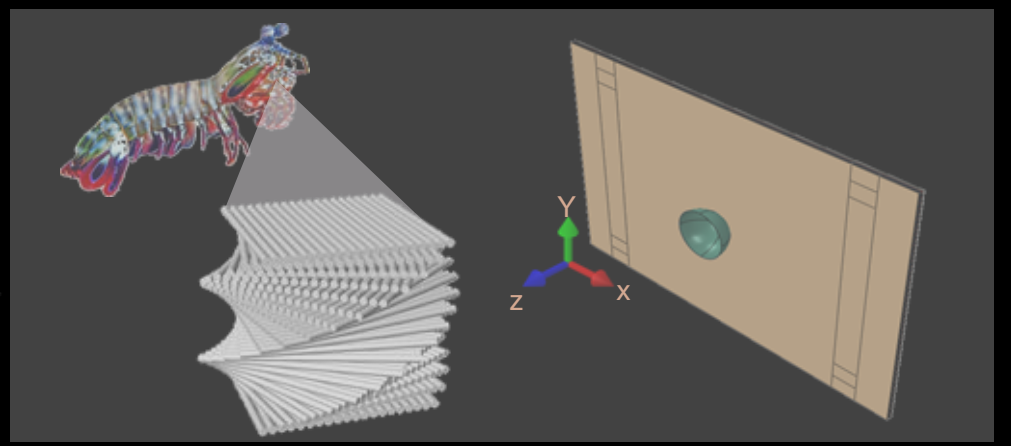
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# Outline

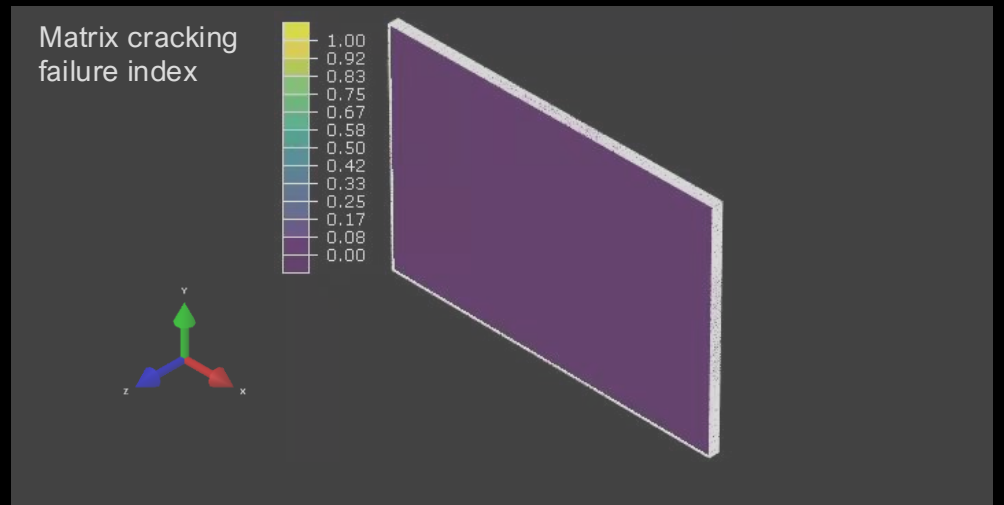
## Background & Motivation



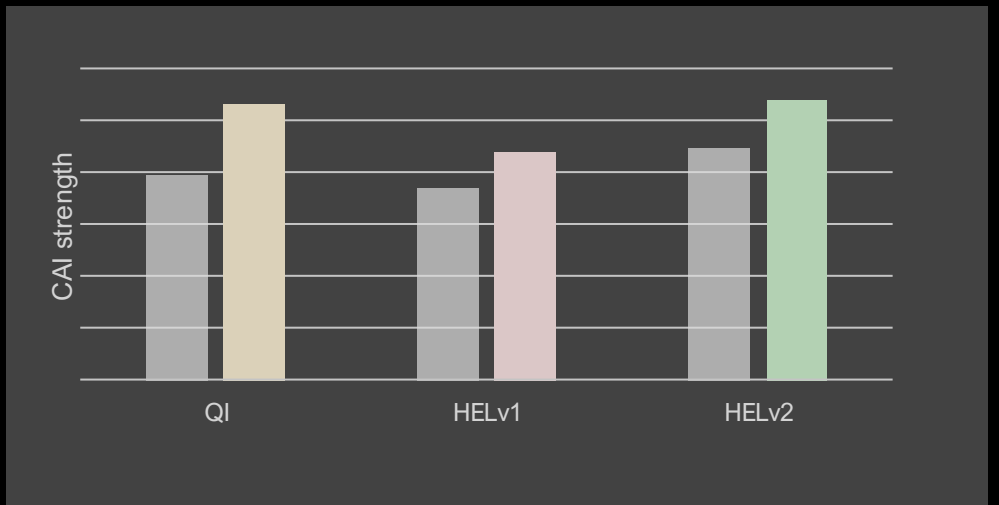

## Design & Methodology




## Results & Discussion




## Summary & Conclusions

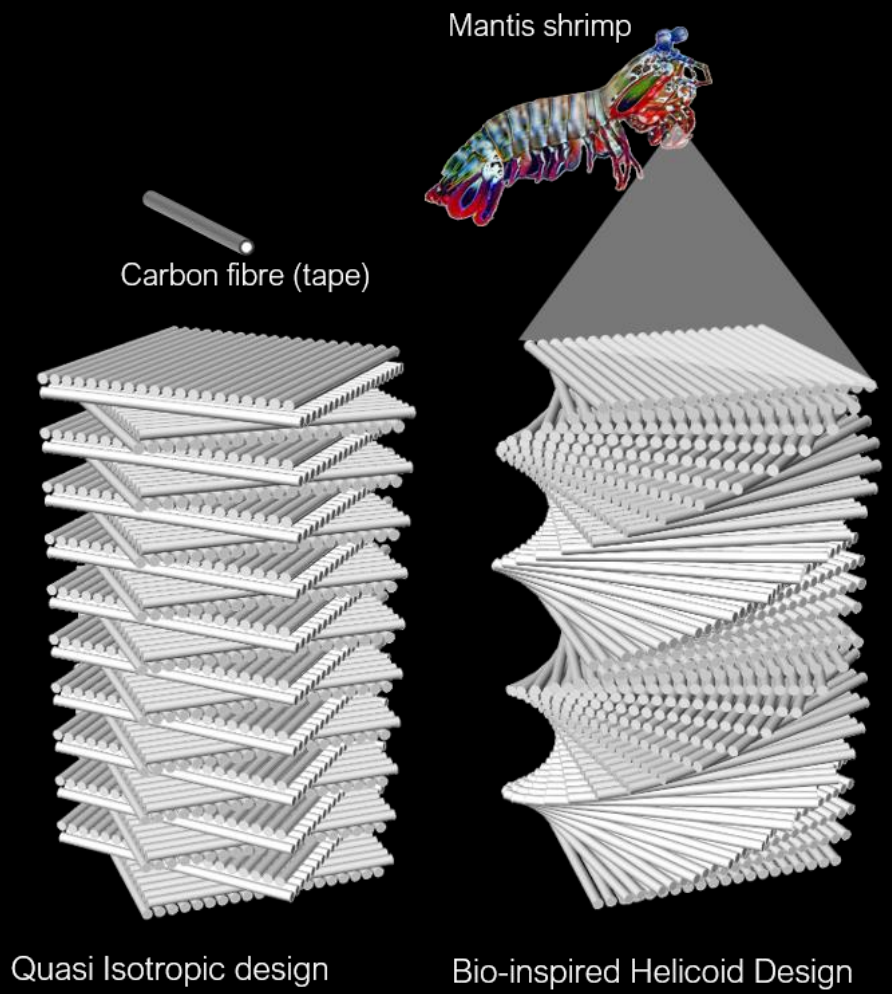


# Design & Methodology

Design concept

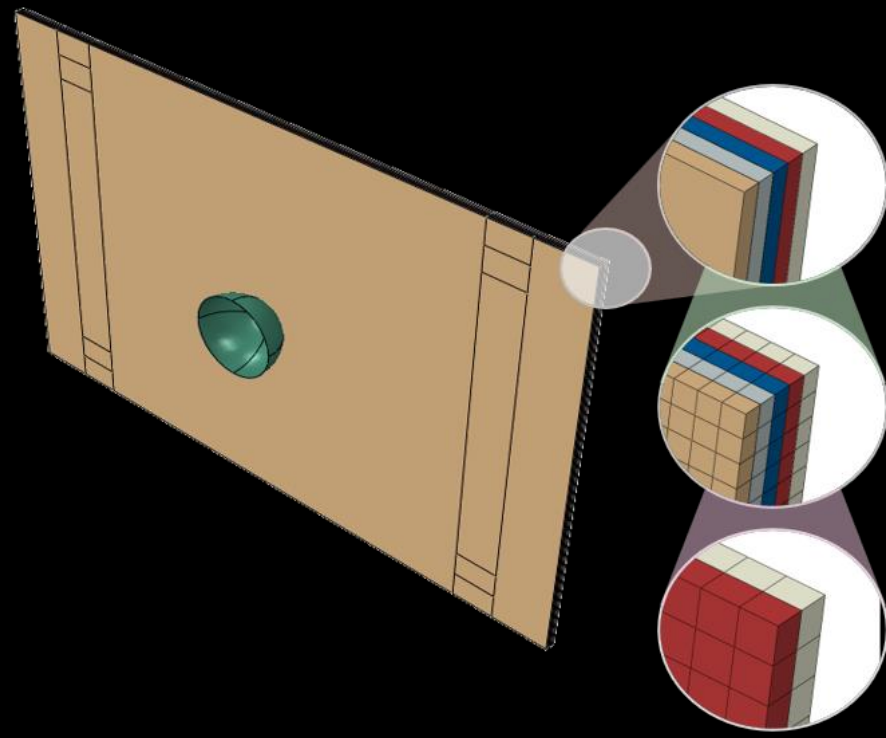
Experimental setup

FEA setup







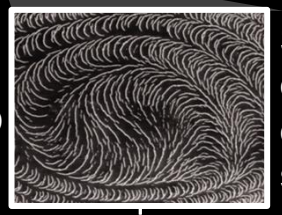


# Design & Methodology

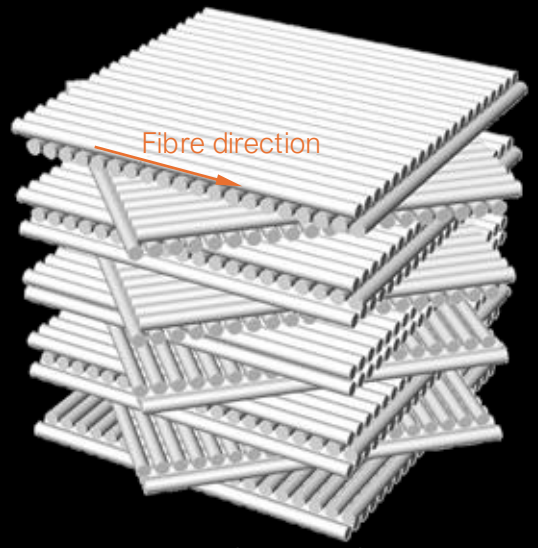
Design concept | Experimental setup | FEA setup

- Baseline (BI) laminates are designed at two thicknesses: thin-wall (2.3 mm) and thick wall (4.4 mm) using conventional 0/90/45 plies.
- The helicoidal concept is inspired by the periodic region of the of the mantis shrimp dactyl club
- The proposed helicoidal layups were designed to have comparable D-matrix to BI laminate.
- HEL-v2 combines a helicoidal core with BI outer plies.



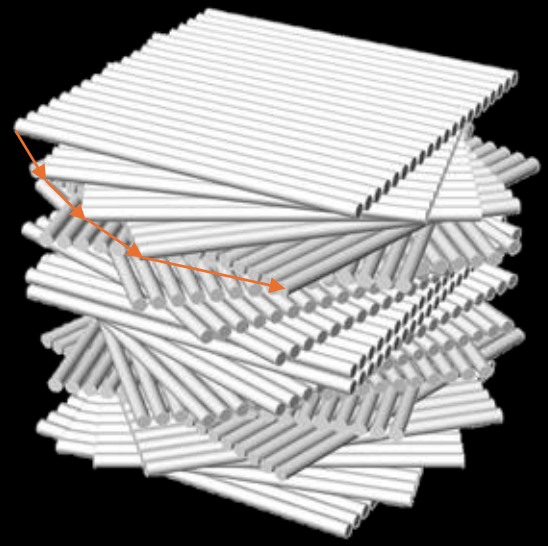


SEM of a damaged coronal cross section [2]



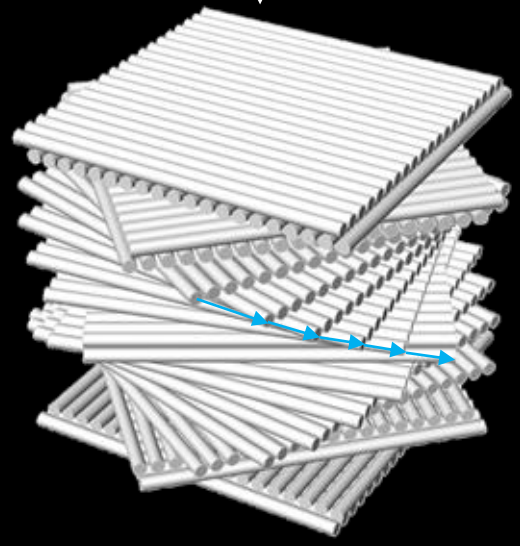

Baseline (BI thin)

$[0/90/\pm45/0/90/0/\pm45/0]_s$



Helicoid v1 (HEL-v1 thin)

$[0/15/30/45/75/-75/-45/-30/-15/0]_s$



Helicoid v2 (HEL-v2 thin)

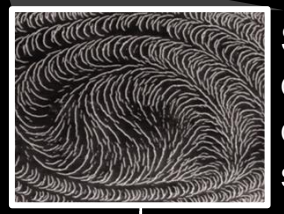
$[0/90/45/-45/-30/-15/-5/5/15/30]_s$

[1] Weaver, J. C., et al. (2012). "The Stomatopod Dactyl Club: A Formidable Damage-Tolerant Biological Hammer." Science 336(6086): 1275-1280.
[2] The ArthroBlogger. (2020, October 13). "Beautiful & terrifying mantis shrimp". Retrieved from https://thearthroblogger.com/2020/10/13/beautiful-terrifying-mantis-shrimp/

# Design & Methodology

Design concept | Experimental setup | FEA setup



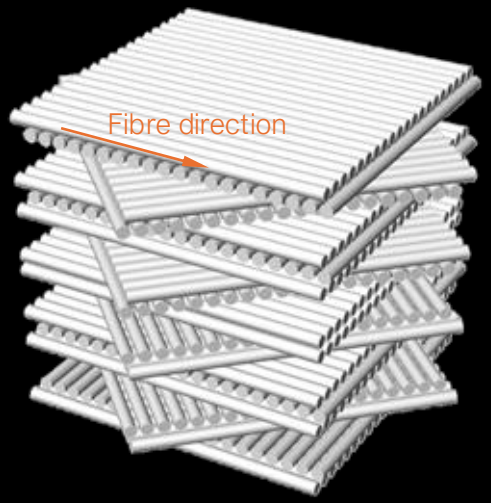


SEM of a damaged coronal cross section [2]

- The thick wall baseline (BI) layup and its helicoidal counterparts are shown below.

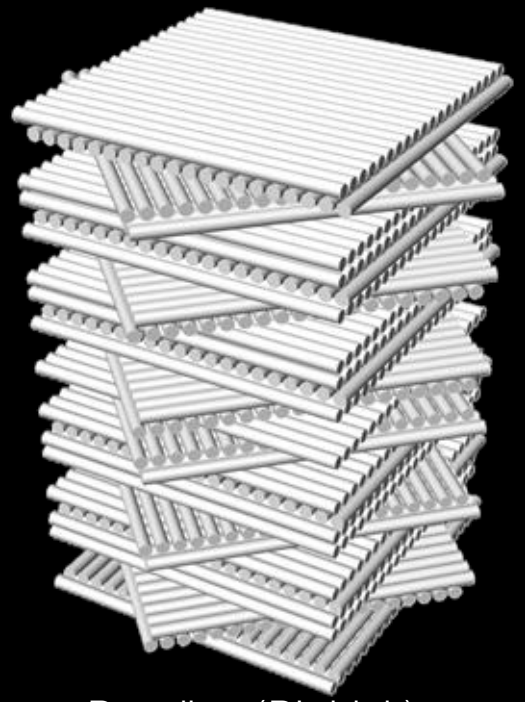
Thick-wall

Thin-wall



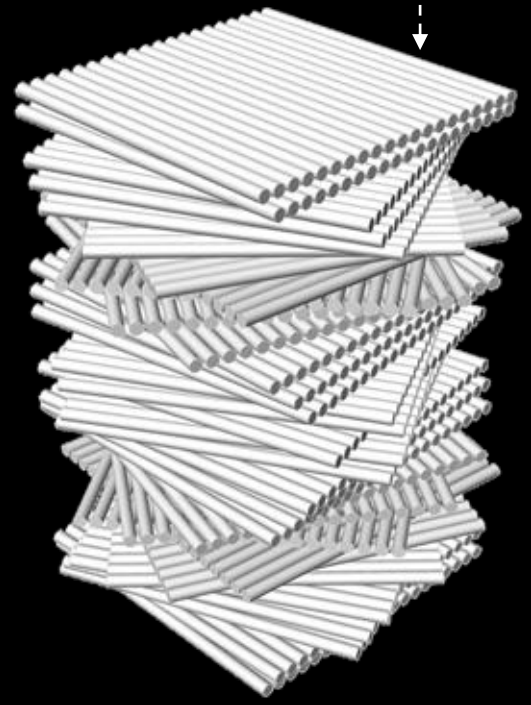

Baseline (BI thin)

$[0/90/\pm45/0/90/0/\pm45/0]_s$



Baseline (BI thick)

$[0/90/\mp45/0_2/90/0/\pm45/0_2/90/0/\pm45/0_{1/2}]_s$



Helicoid v1 (HEL-v1 thick)

$[-16/-13/6/9/14/31/50/69/-69/-50/-31/-14/-9/-6/13/16/0_{1/2}]_s$



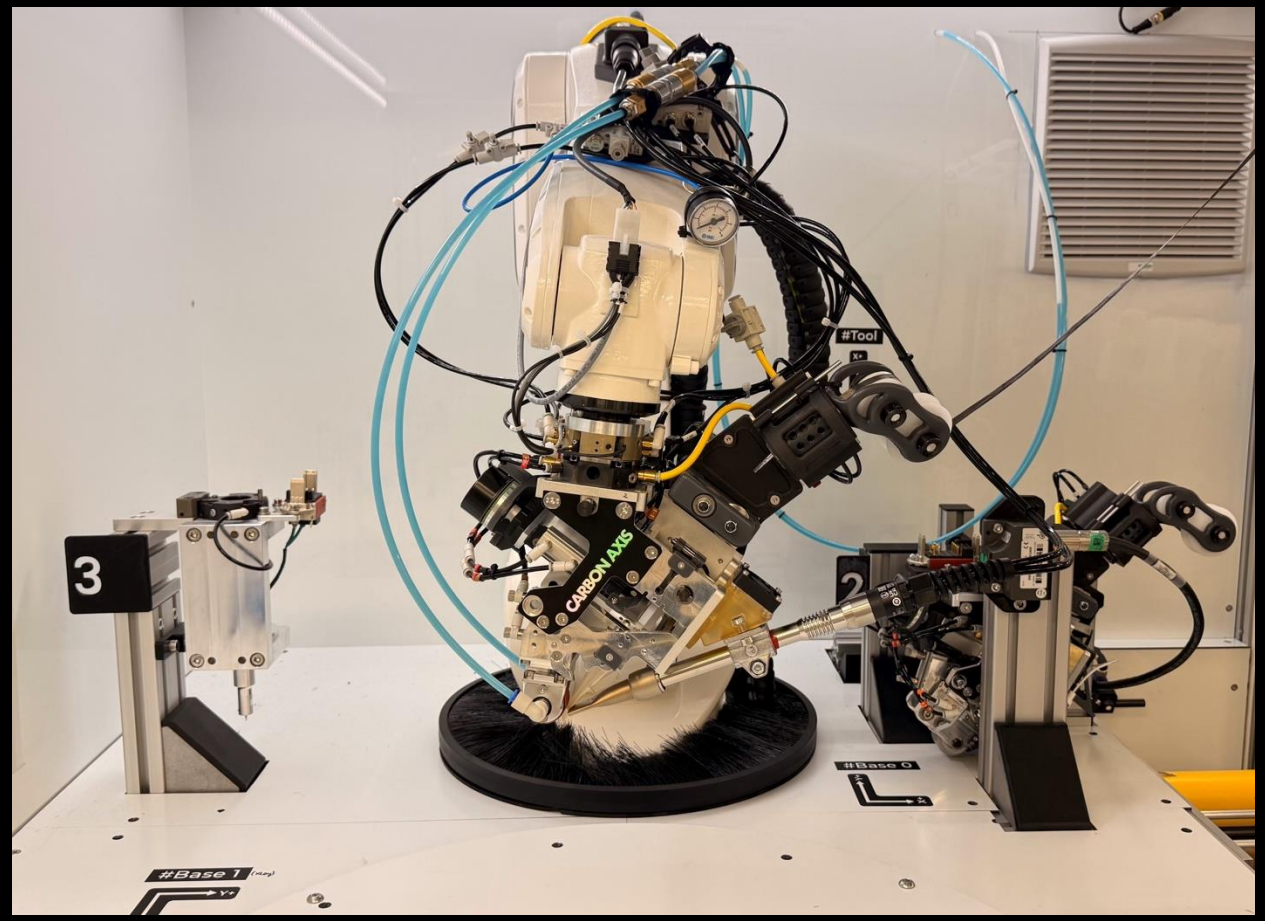
Helicoid v2 (HEL-v2 thick)

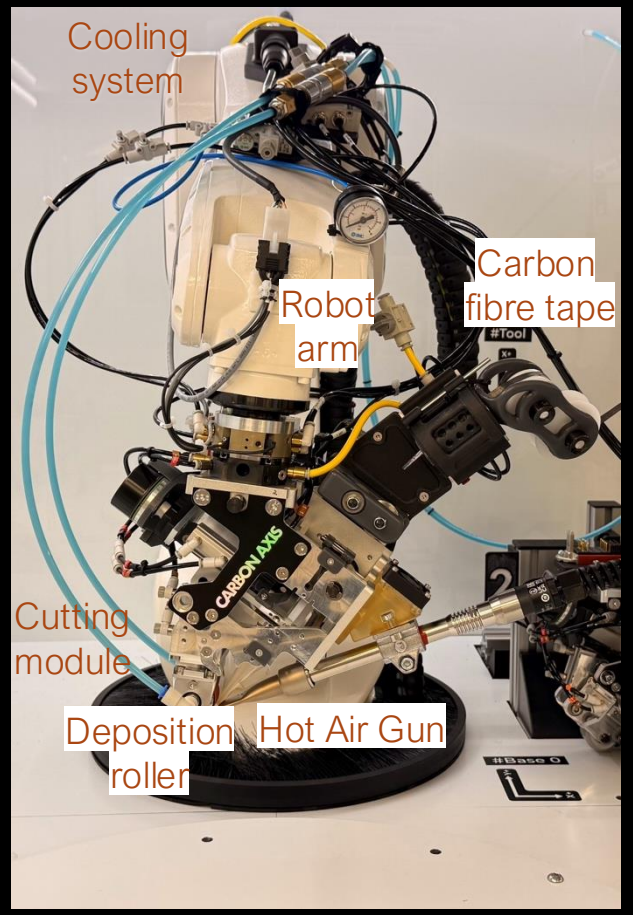
$[0/90/45/-45/-33/-13/7/27/37/42/-42/-37/-27/-7/13/33/0_{1/2}]_s$

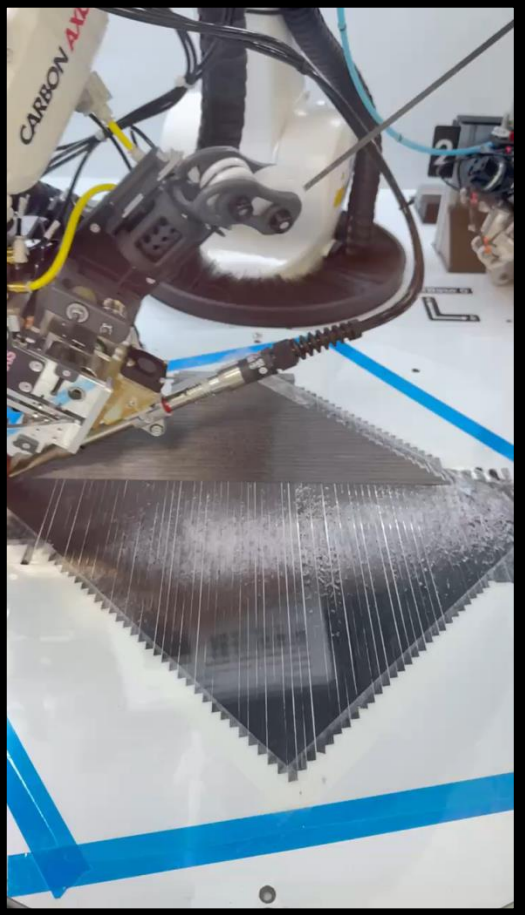
# Design & Methodology

Design concept | Experimental setup | FEA setup

- Key enabler of this work - the Automated Fibre Placement (AFP) machine



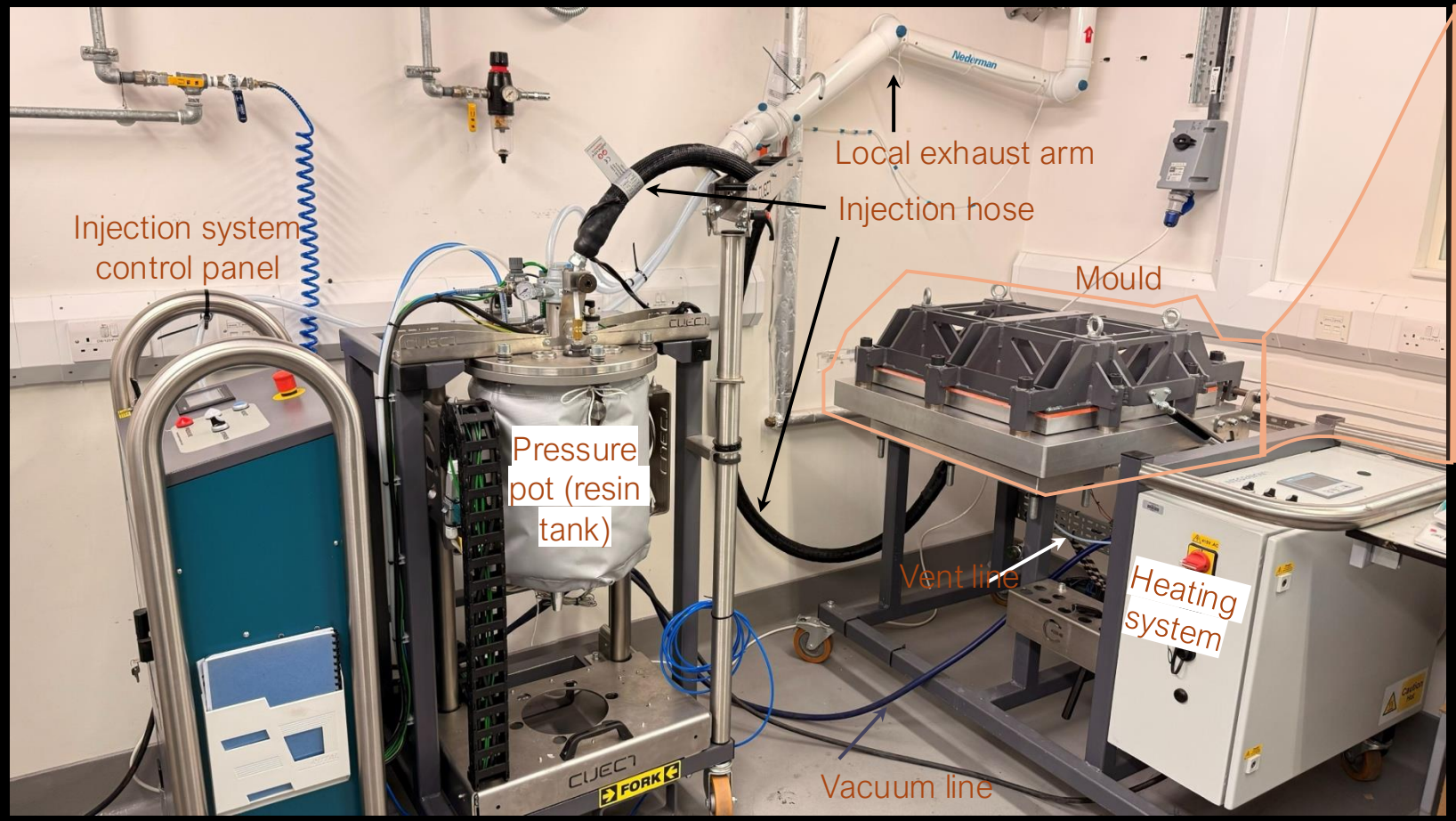



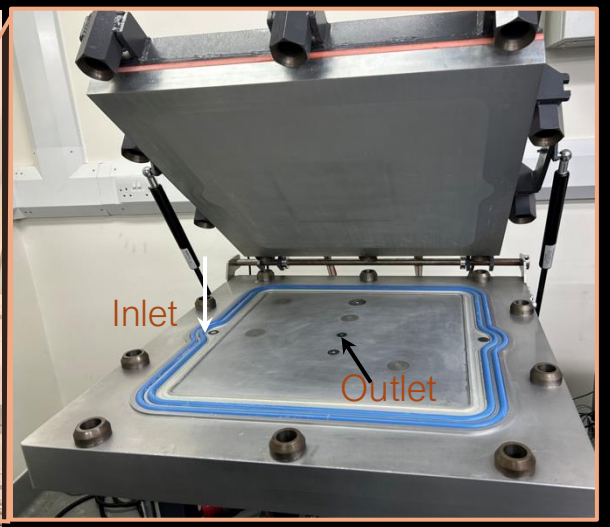



# Design & Methodology

Design concept | Experimental setup | FEA setup

- Key enabler of this work - the Resin Transfer Moulding (RTM) machine



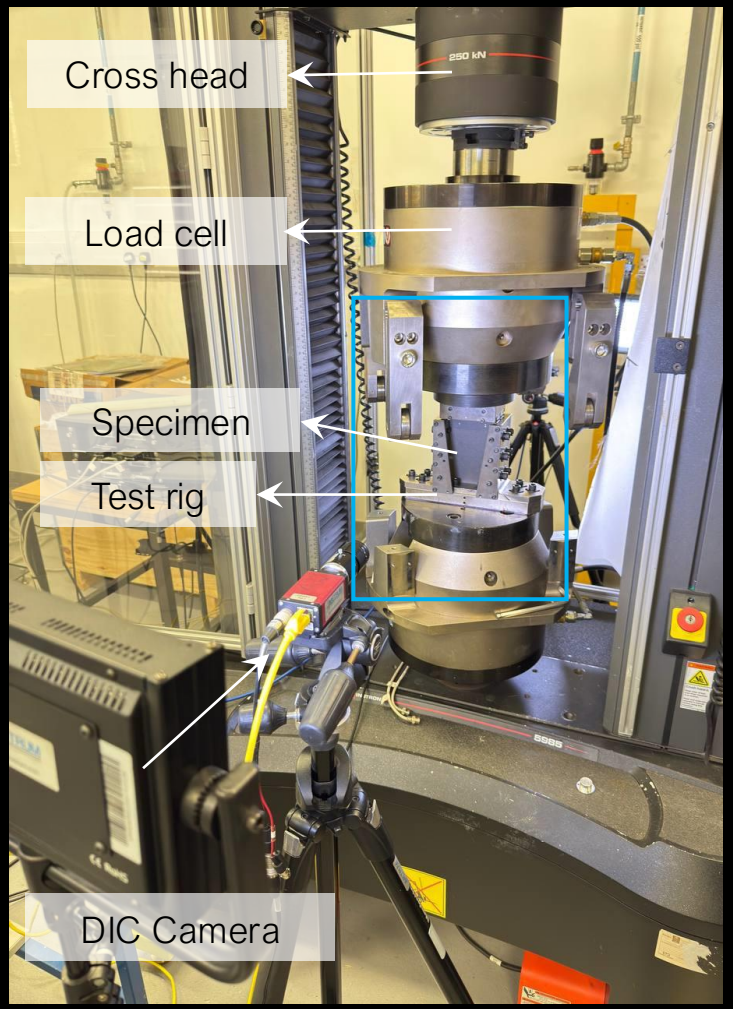

Mould

# Design & Methodology

Design concept | Experimental setup | FEA setup




LVI testing setup

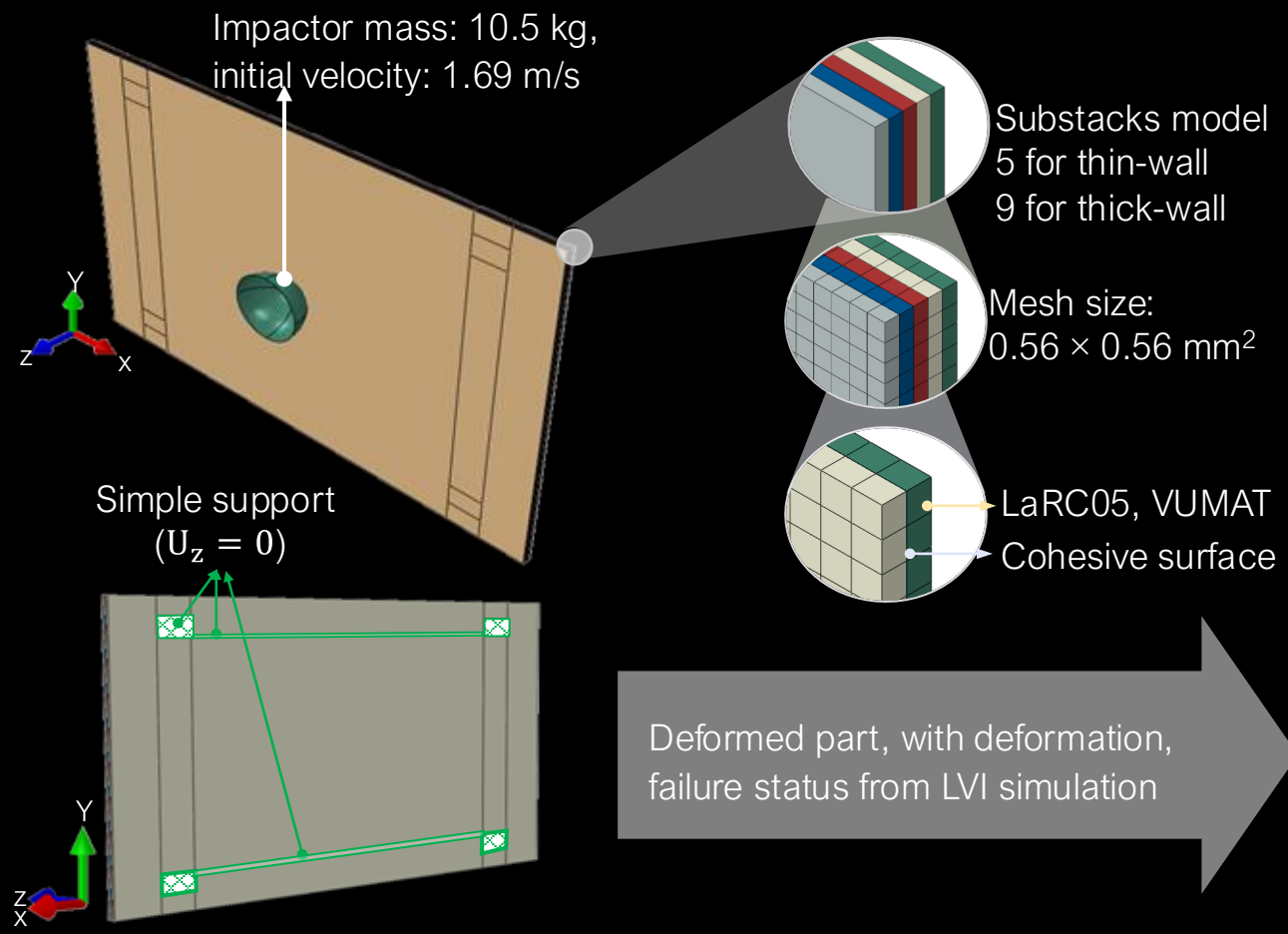



CAI testing setup

| Material | IM7/8552 |
|---|---|
| Specimen dimensions | 150 mm x 100 mm |
| **Low-velocity impact (LVI)** | |
| Test standard | ASTM D7136 |
| Impactor diameter | 20 mm |
| Impactor mass | 10.5 kg |
| Impact energy | 15 J for thin-wall<br>25 J for thick-wall |
| Experimentally-measured variables | Time (ms)<br>Force (kN) |
| Sampling frequency | 500 kHz |
| **Compression after impact (CAI)** | |
| Test standard | ASTM D7137 |
| Specimen dimensions | 150 mm x 100 mm x 2.3 mm |
| Loading rate | 1 mm/min |
| Experimentally-measured variables | • Crosshead displacement<br>• Force (kN) |

# Design & Methodology

Design concept | Experimental setup | FEA setup




- Abaqus/Explicit simulations
- LaRC05 damage model implemented through a VUMAT subroutine [1]
- Failure modes captured: matrix cracking, matrix splitting, fibre tension, fibre kinking

- LVI and CAI boundary conditions (BCs) follow the respective ASTM standards
- Only the specimen is modelled; testing rigs are excluded

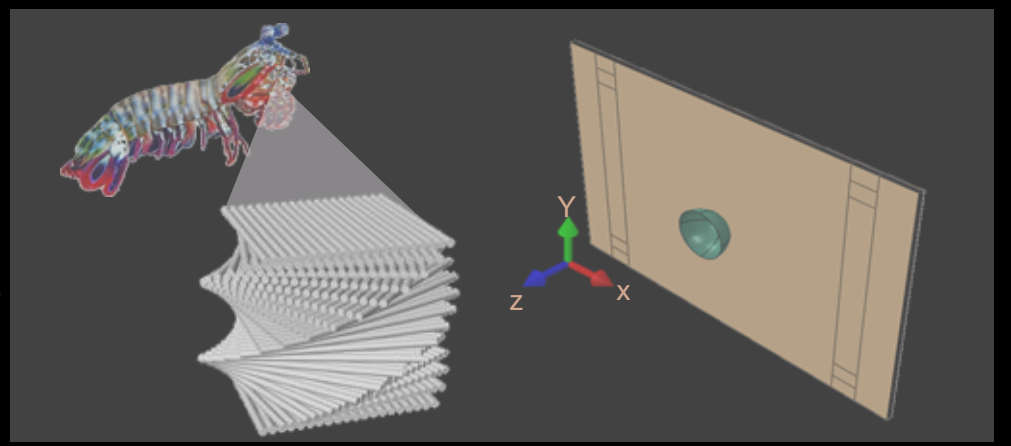
[1] ST Pinho et al. Material and structural response of polymer-matrix fibre-reinforced composites. Journal of Composite Materials. 46, 2313–2341, 2012

# Outline

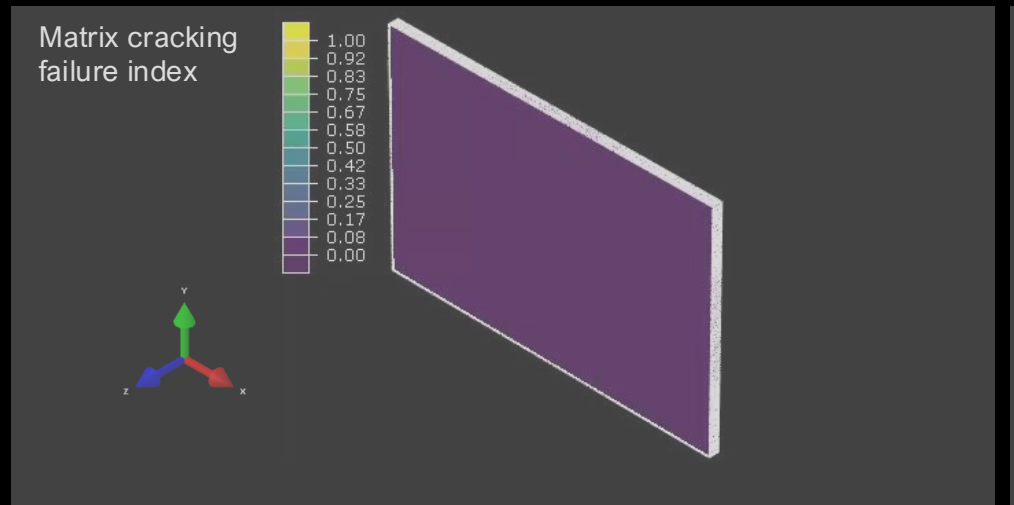
## Background & Motivation



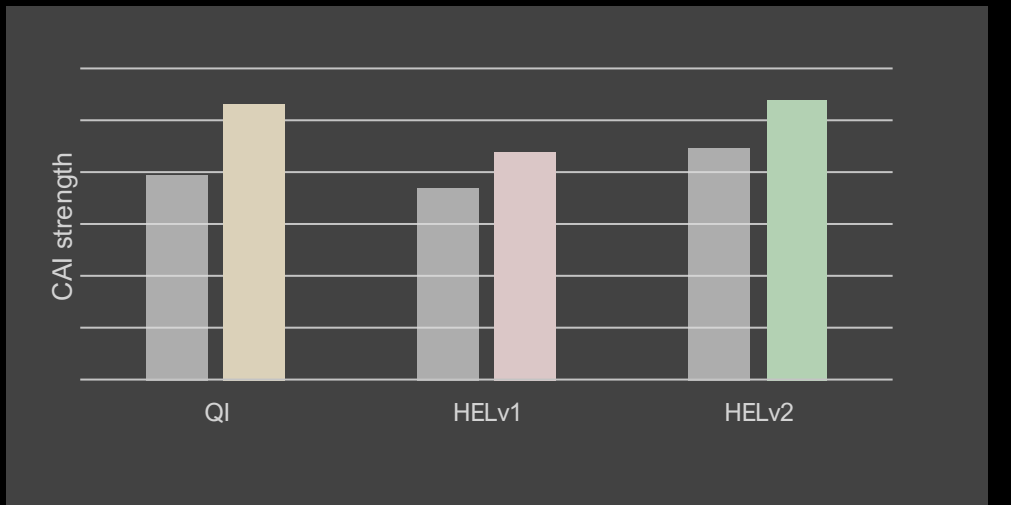

## Design & Methodology




## Results & Discussion




## Summary & Conclusions



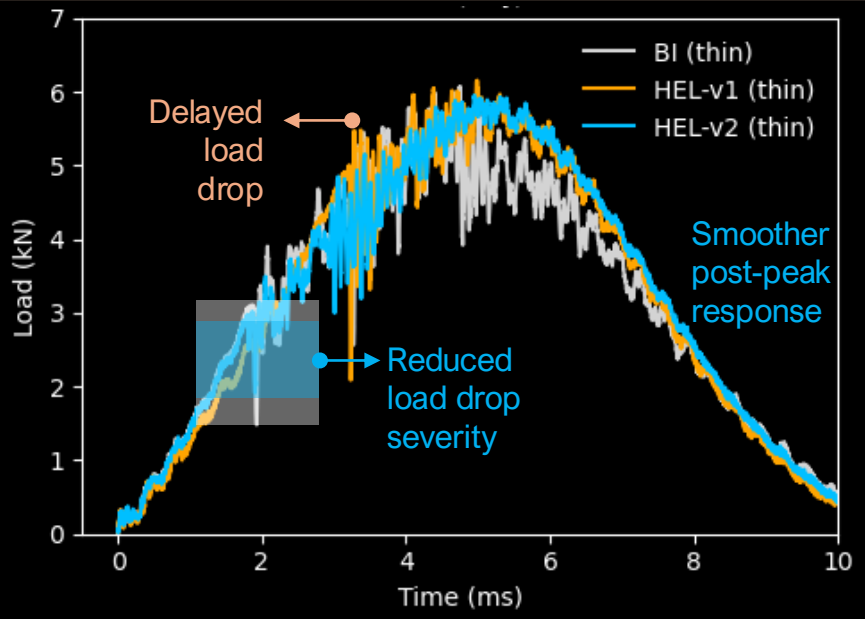
# Results & Discussion

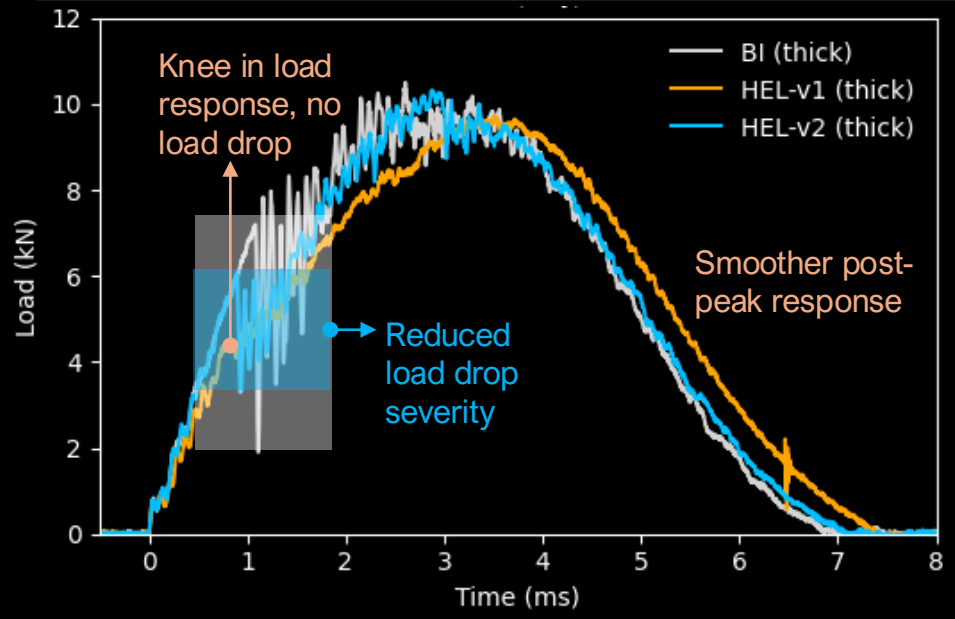
Experimental results | FEA validation | Damage analysis

## Thin-wall

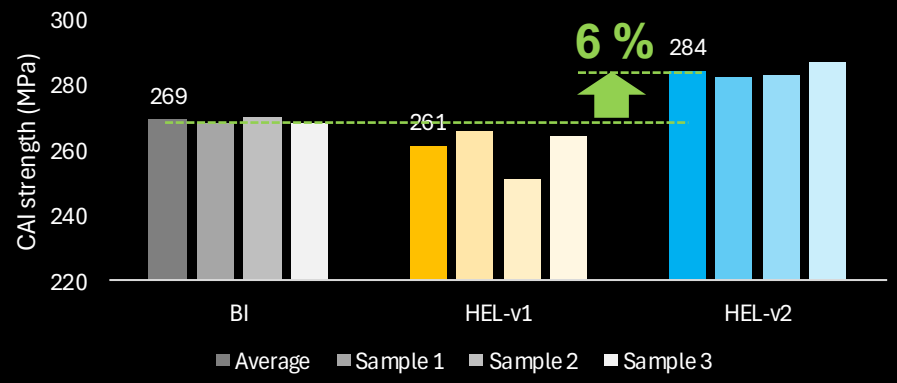
## Thick-wall

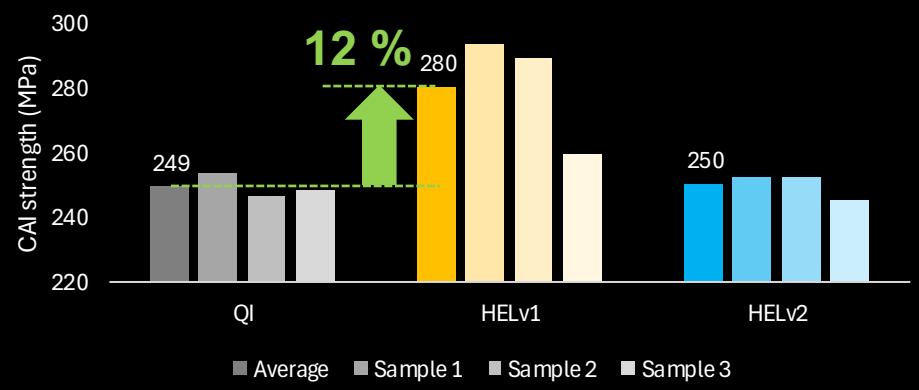
LVI




CAI

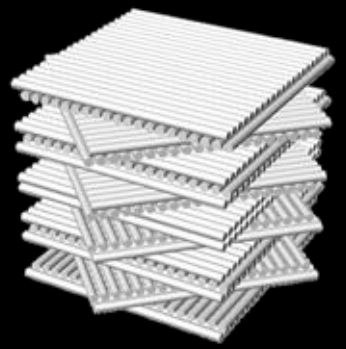


# Results & Discussion

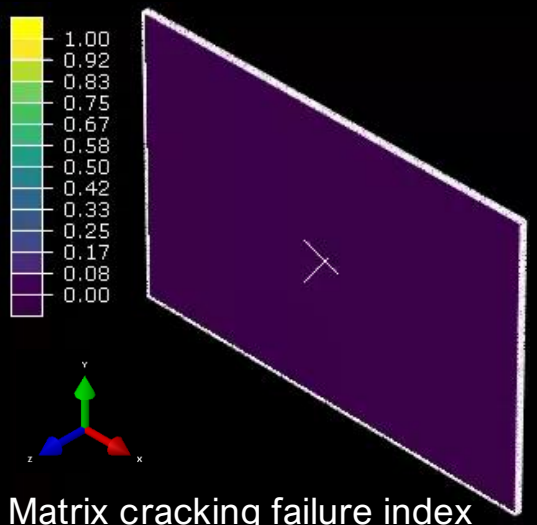
Experimental results | FEA validation | LVI | Damage analysis

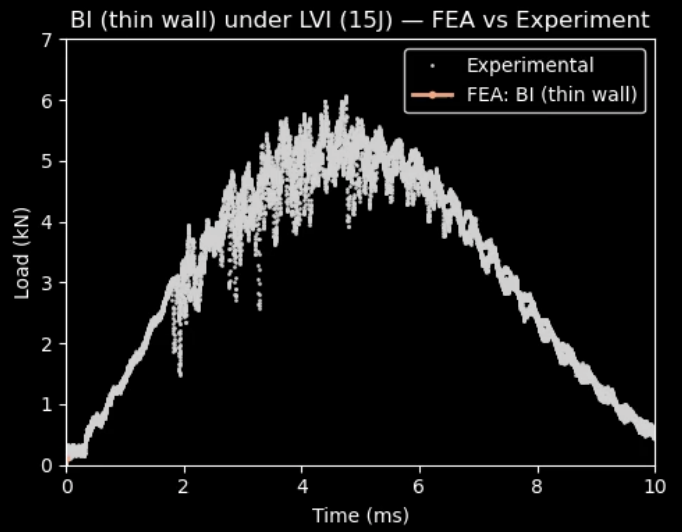
Thin-wall



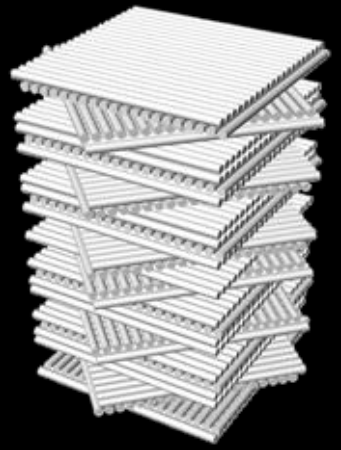
BI

Matrix cracking failure index



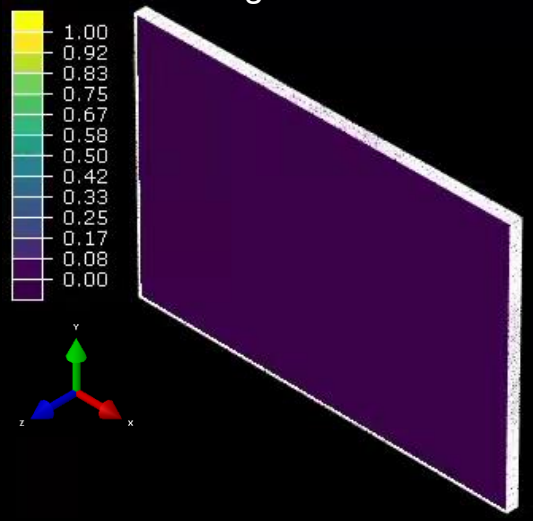



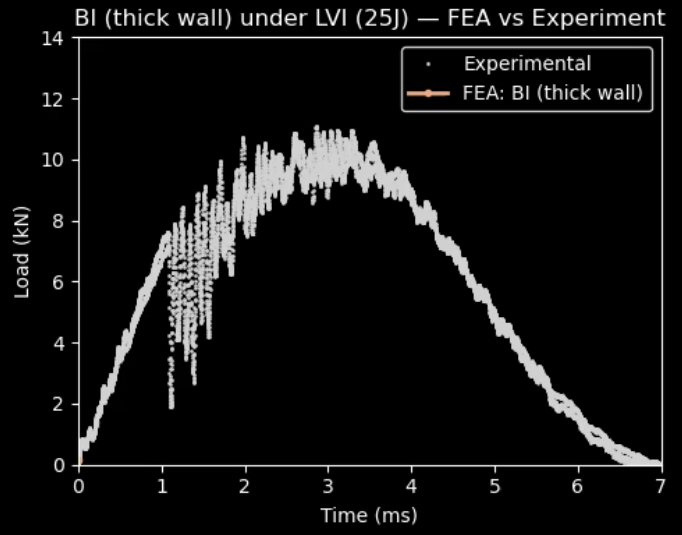

Thick-wall



BI

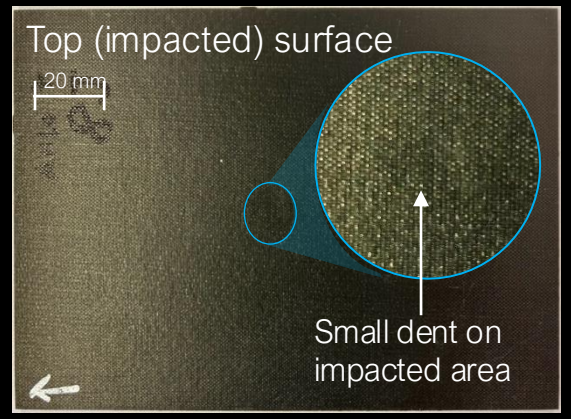
Matrix cracking failure index

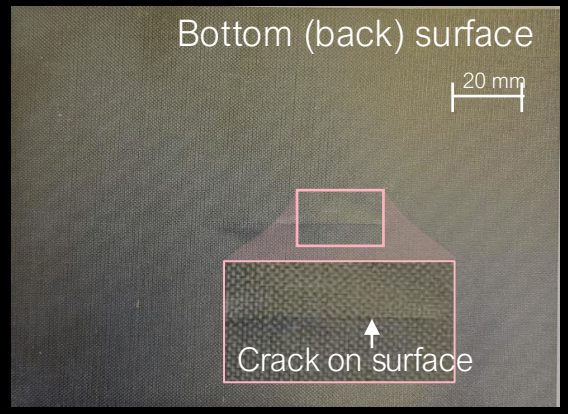


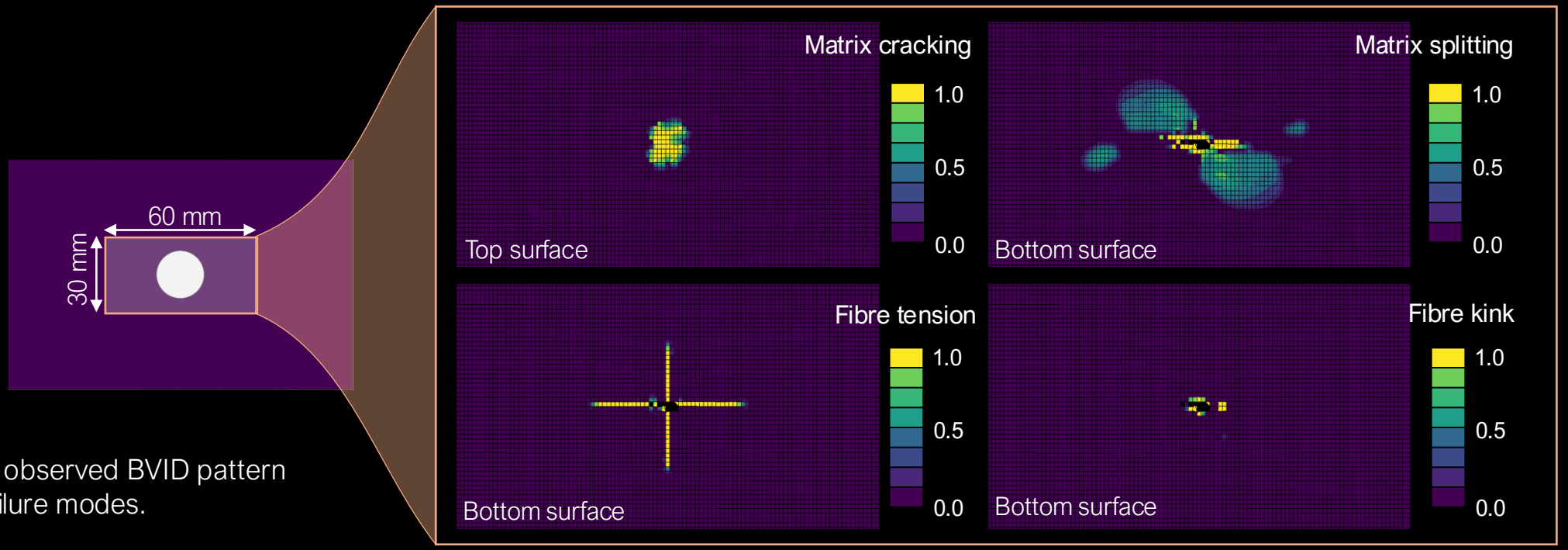



# Results & Discussion

Experimental results | FEA validation | LVI | Damage analysis



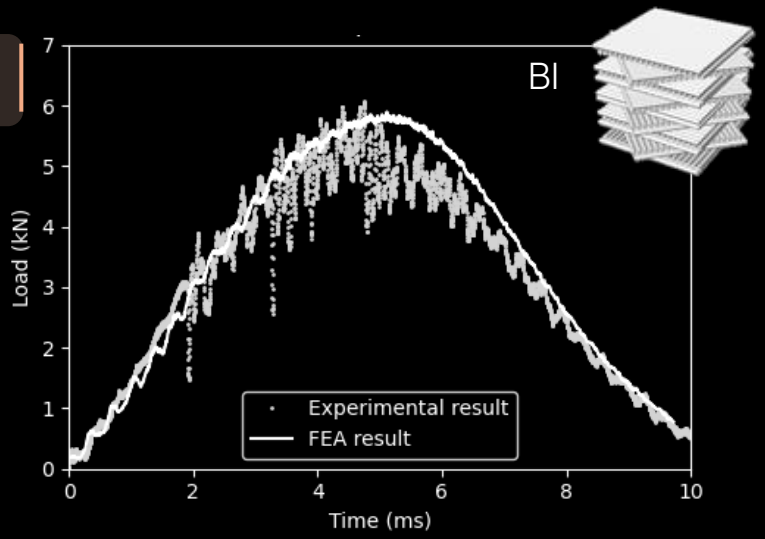



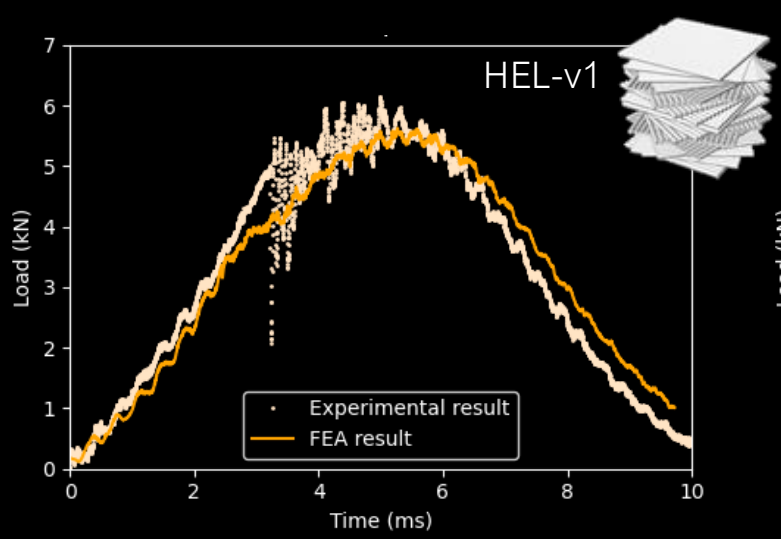

FEA reproduces the observed BVID pattern and the dominant failure modes.

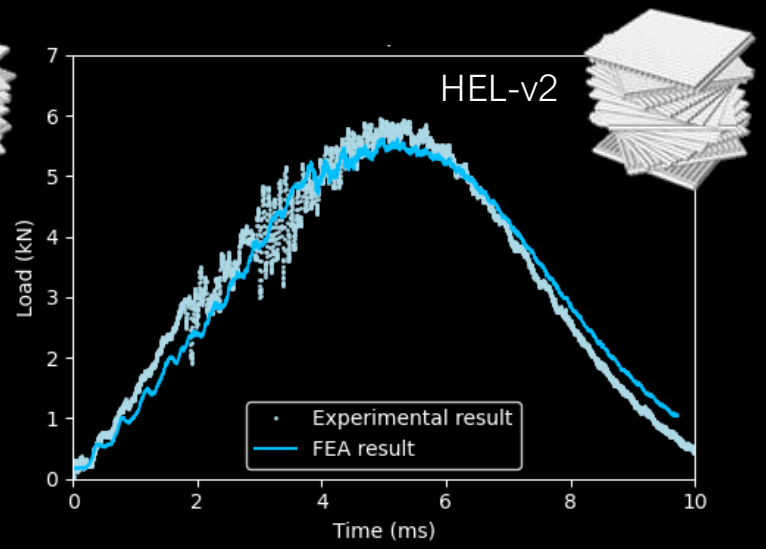
# Results & Discussion

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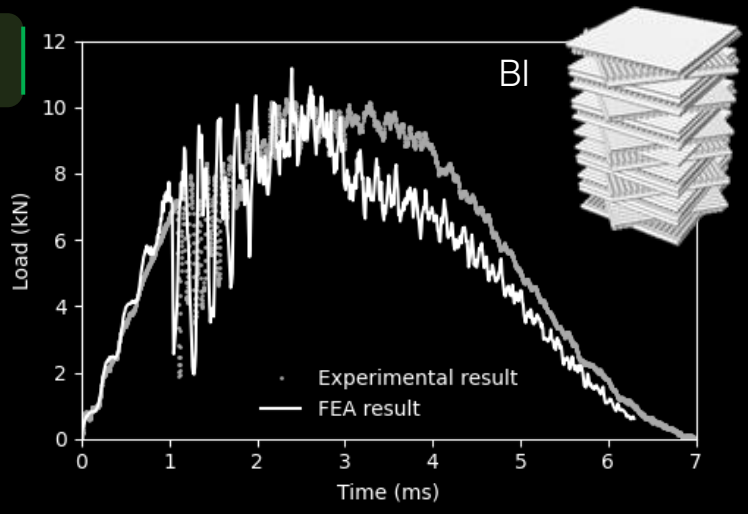
Thin-wall

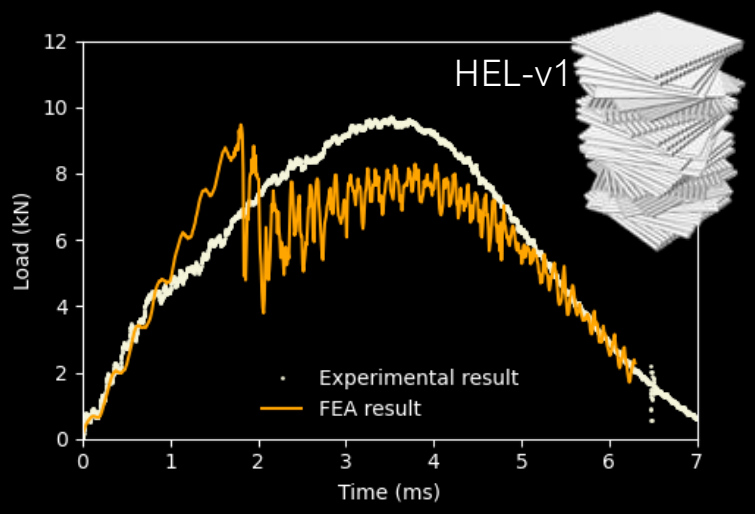


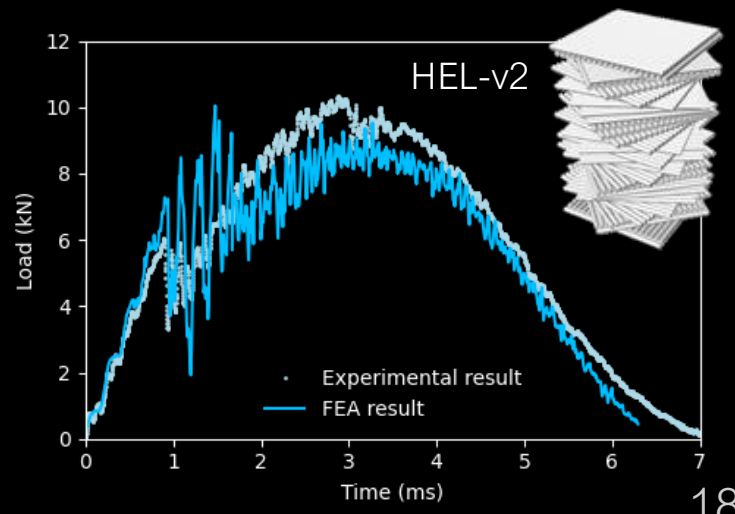




Thick-wall

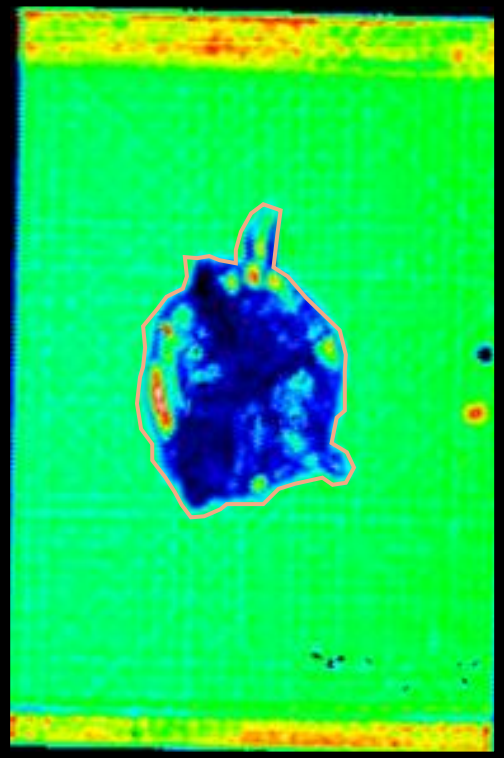


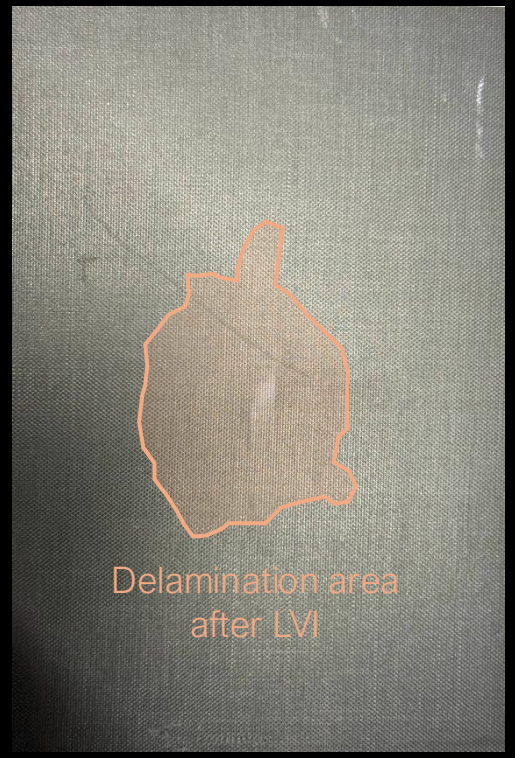



# Results & Discussion

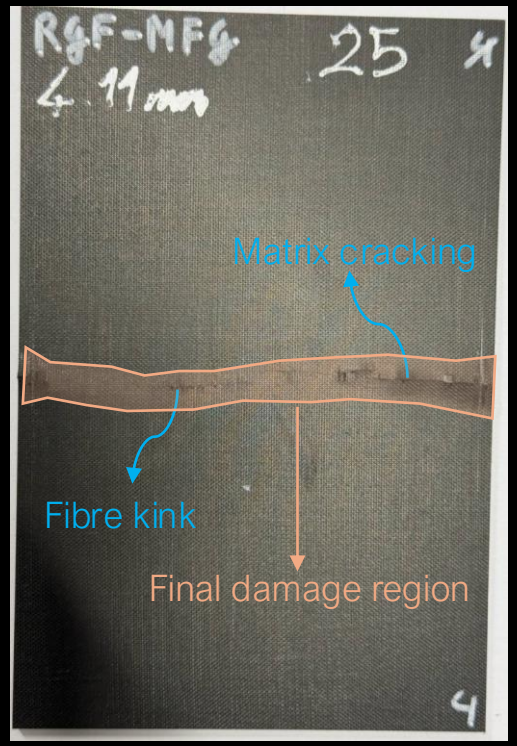
Experimental results | FEA validation | CAI | Damage analysis

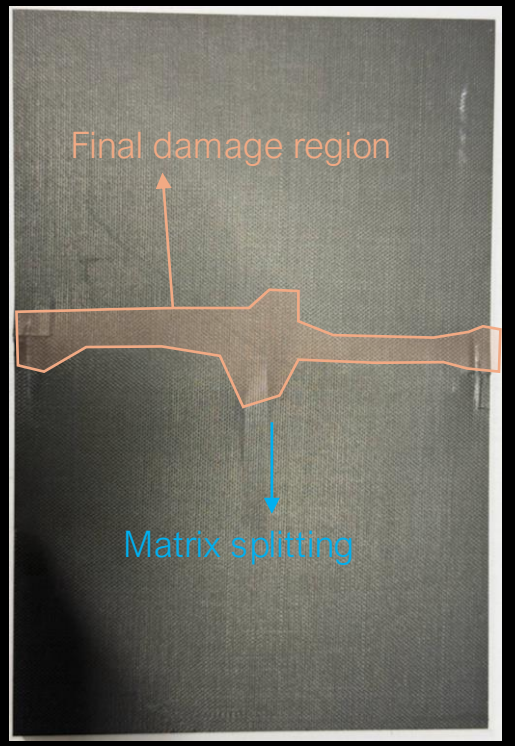





C-Scan of the BI (thick) specimen after LVI (25J)

Transferred to CAI

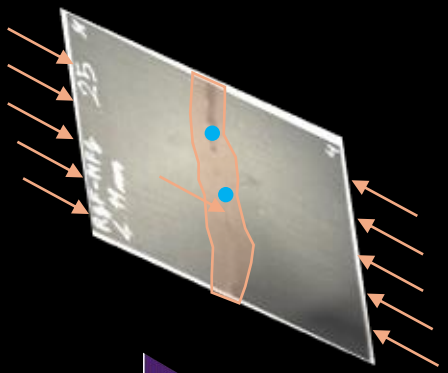


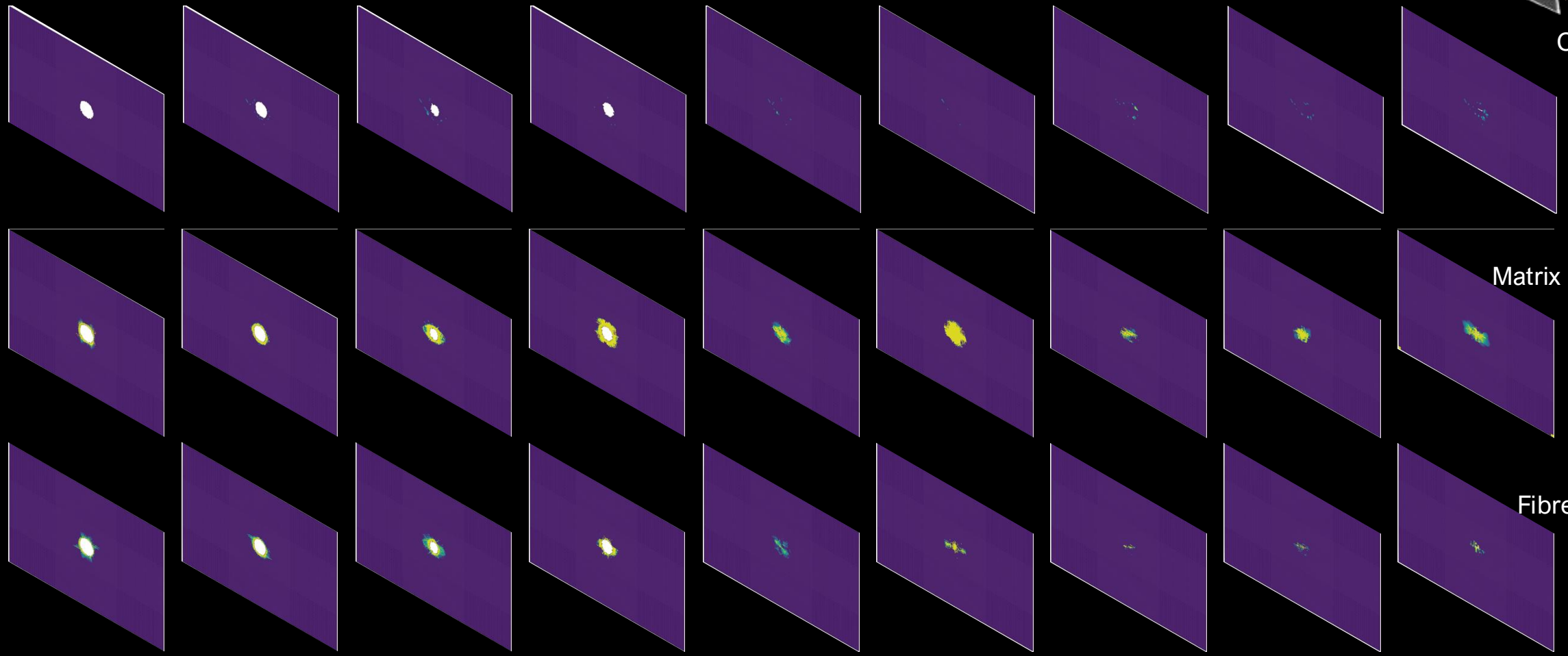




Failed specimen after CAI

# Results & Discussion

Experimental results | FEA validation | CAI | Damage analysis

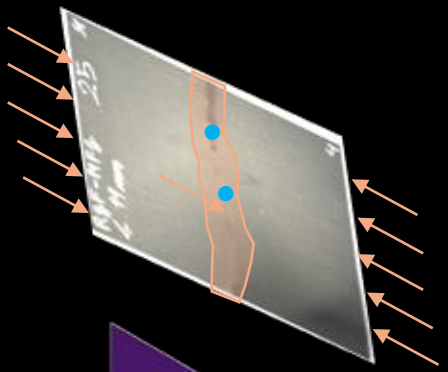


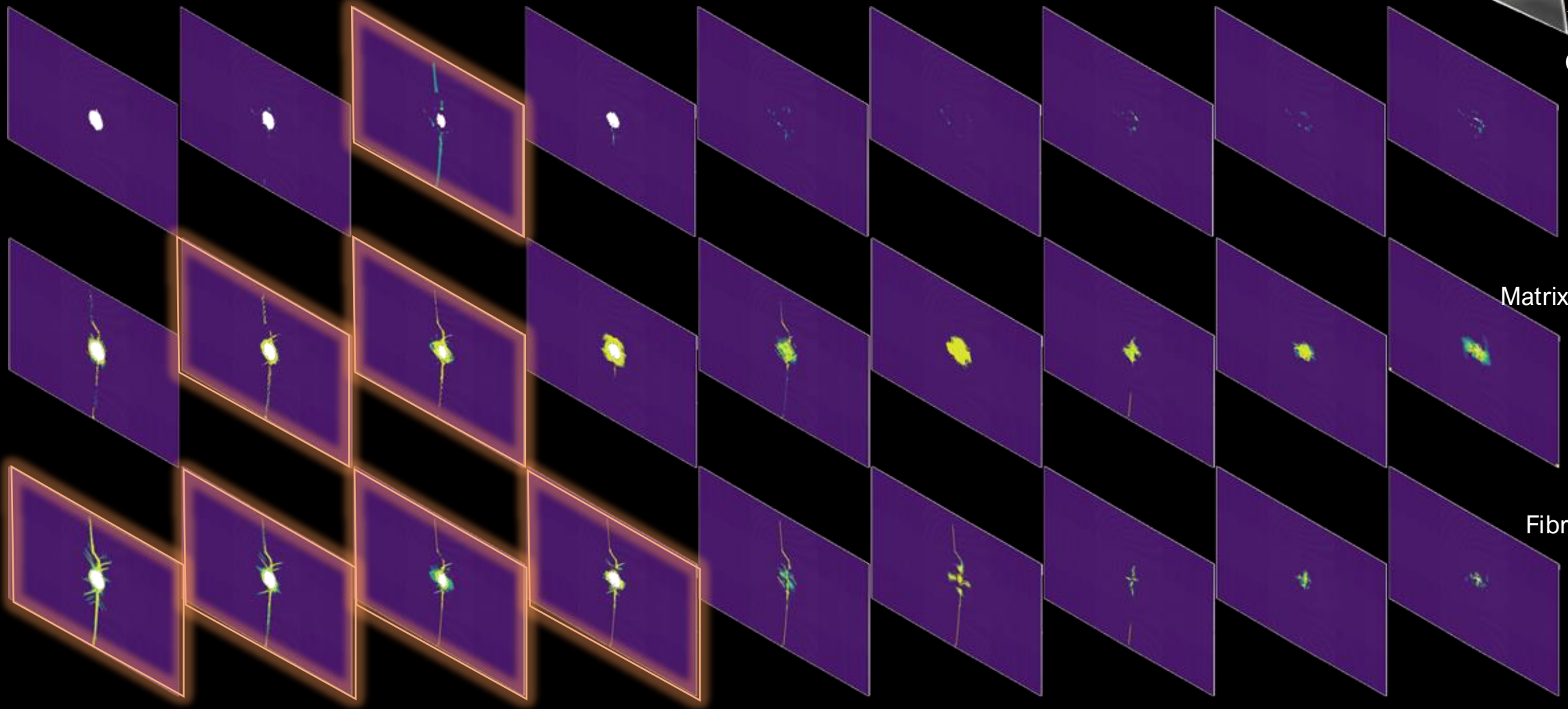



* The FEA-predicted damage distribution of a BI (thick) specimen under CAI

# Results & Discussion

Experimental results | FEA validation | CAI | Damage analysis

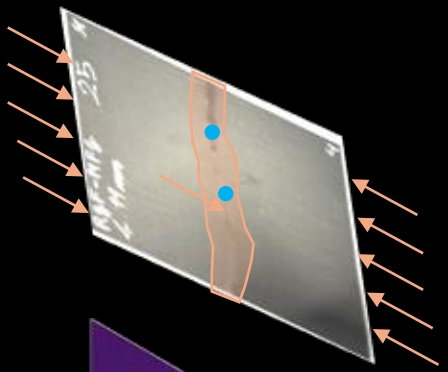


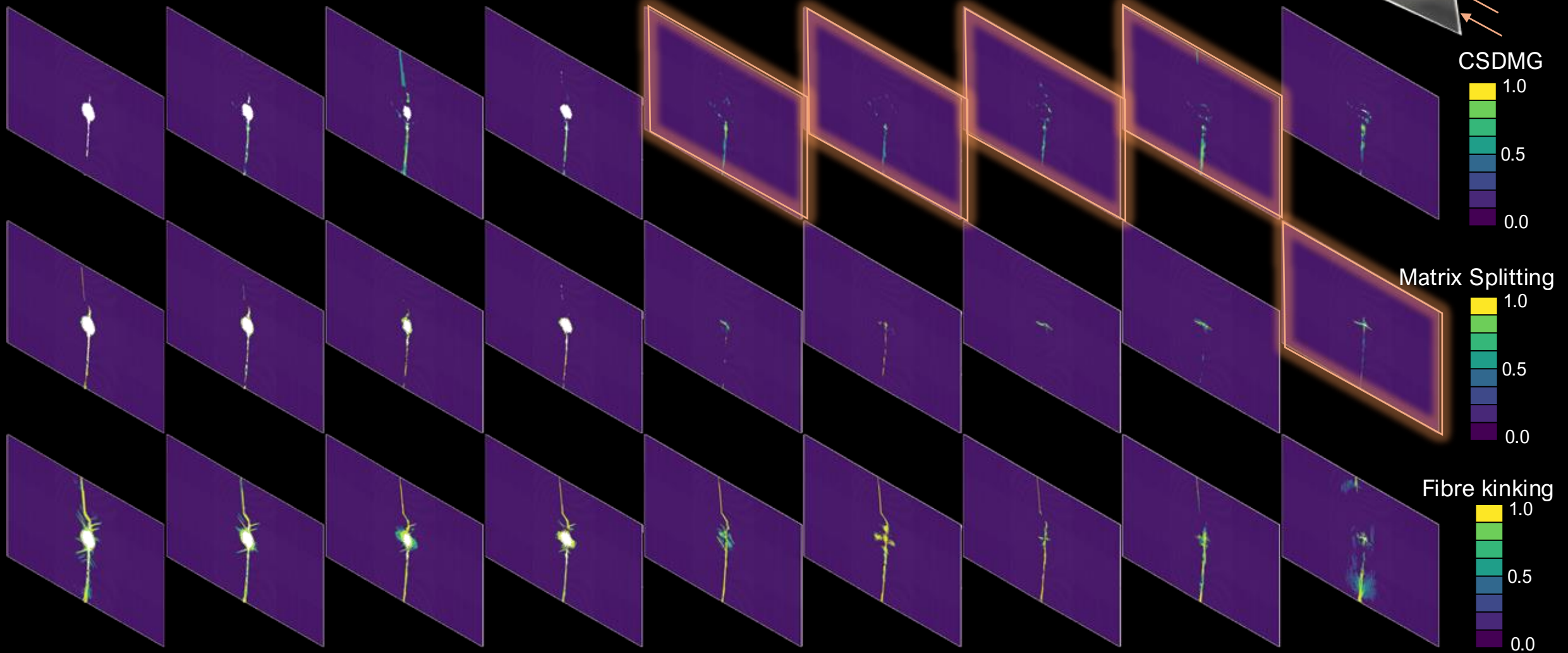



* The FEA-predicted damage distribution of a BI (thick) specimen under CAI, at a compressive displacement at 0.752 mm

# Results & Discussion

Experimental results | FEA validation | CAI | Damage analysis

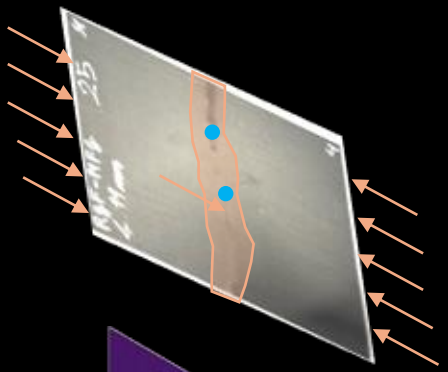


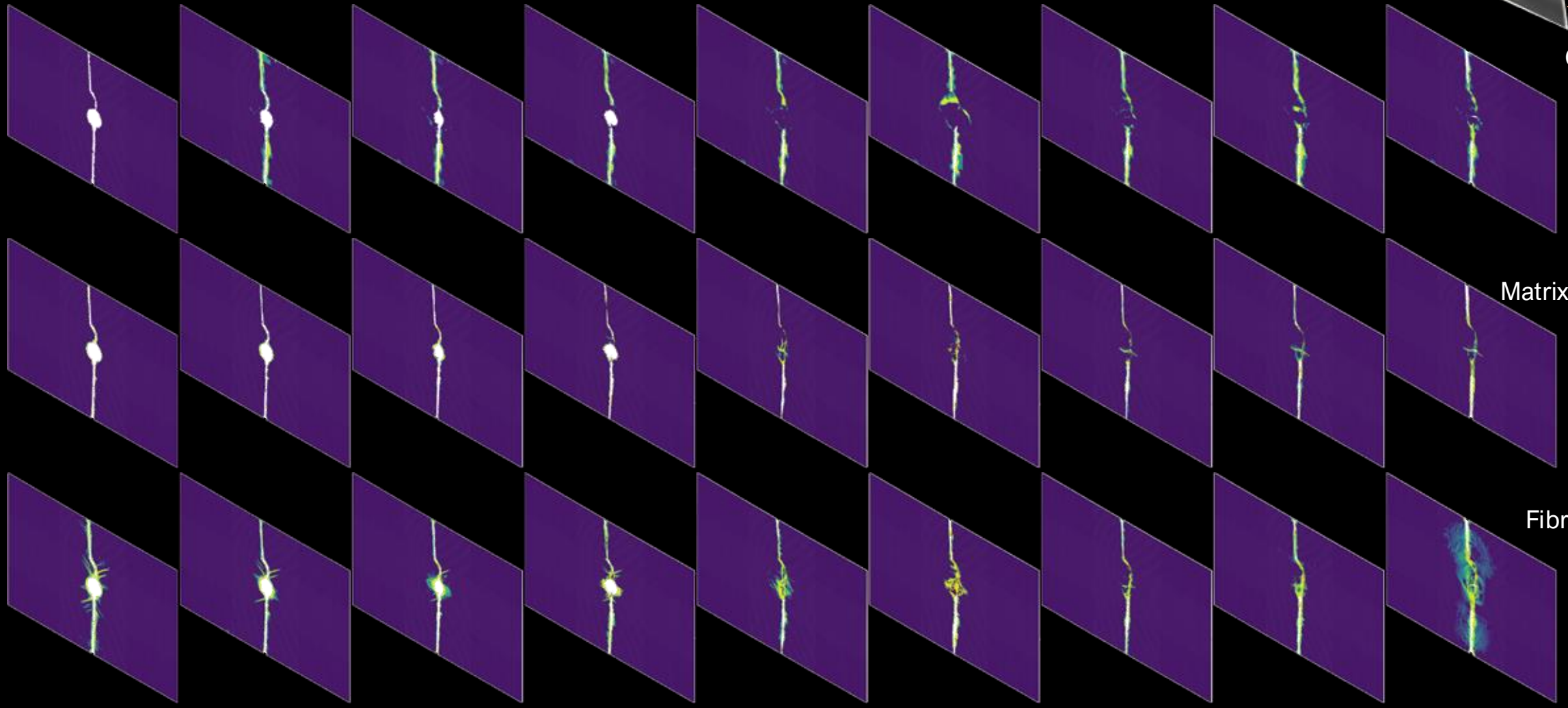



* The FEA-predicted damage distribution of a BI (thick) specimen under CAI, at a compressive displacement at 0.756 mm

# Results & Discussion

Experimental results | FEA validation | CAI | Damage analysis

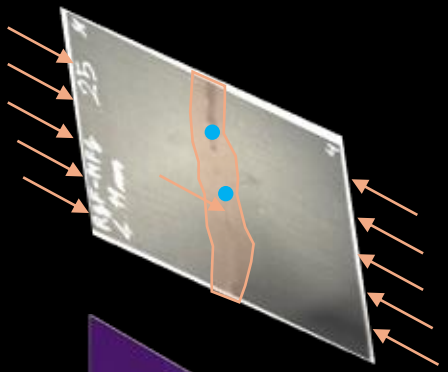


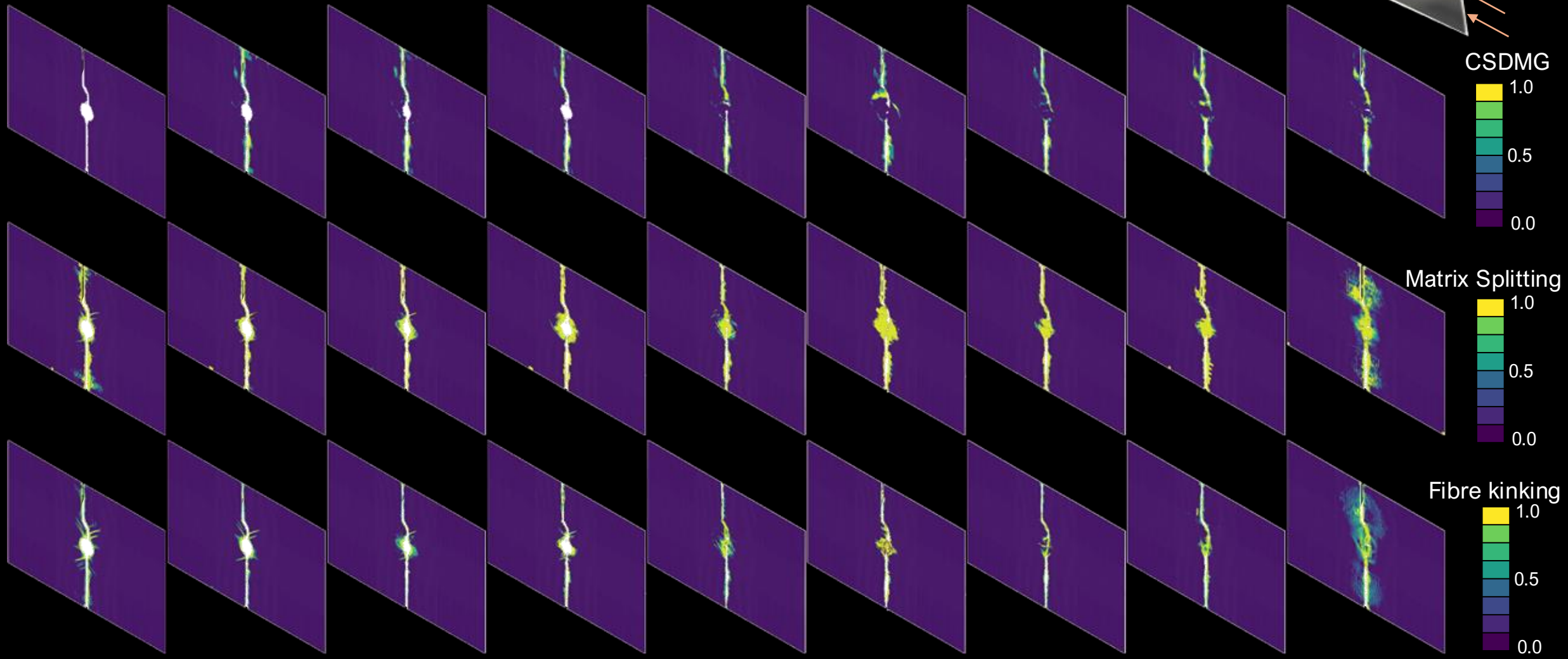



* The FEA-predicted damage distribution of a BI (thick) specimen under 25J CAI, at a compressive displacement at 0.764 mm

# Results & Discussion

Experimental results | FEA validation | CAI | Damage analysis

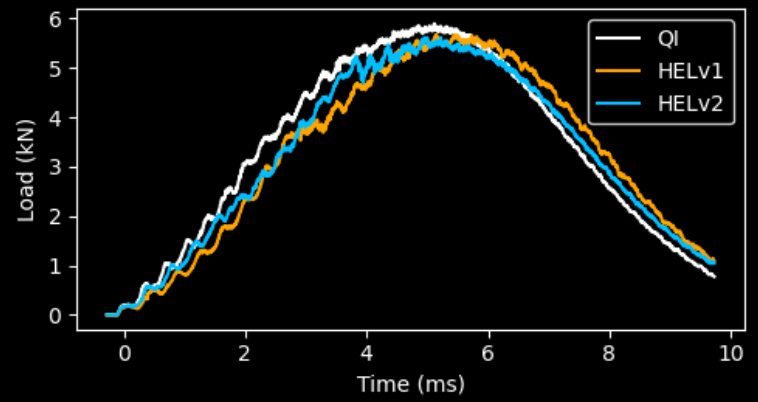


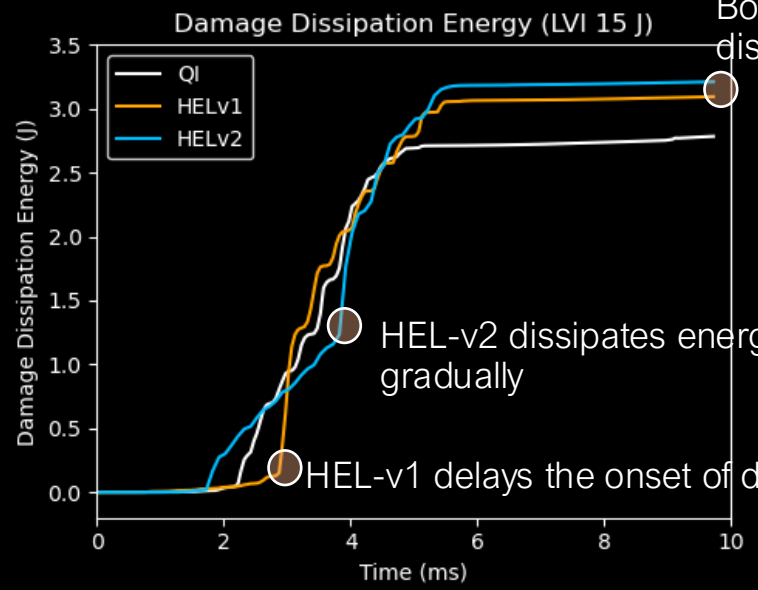



* The FEA-predicted damage distribution of a BI (thick) specimen under 25J CAI, at a compressive displacement at 0.768 mm

# Results & Discussion

Experimental results | FEA validation | Damage analysis




Both Helicoidal designs dissipate more energy

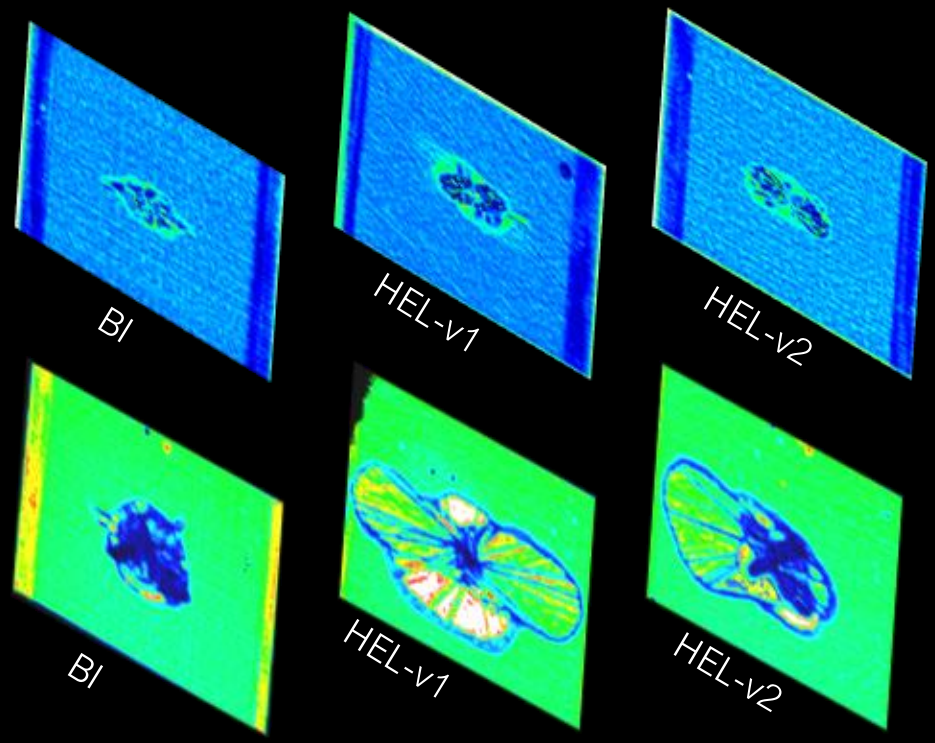
HEL-v2 dissipates energy more gradually

HEL-v1 delays the onset of damage

* FEA results

Thin-wall

Thick-wall



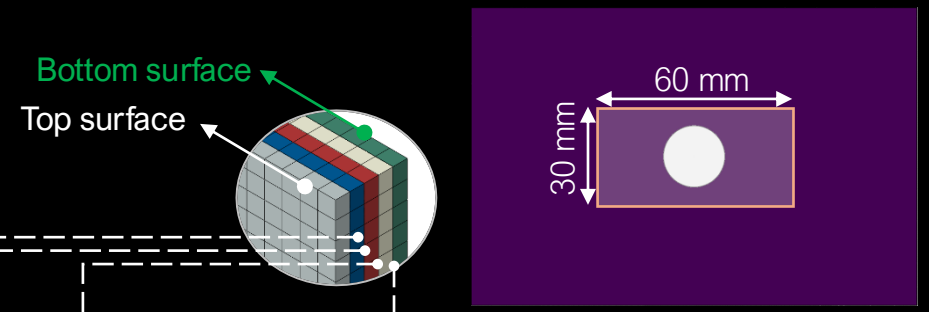

Distributing damage over a wider area while reducing its local intensity may enhance post-impact load-bearing capability and damage tolerance [1]

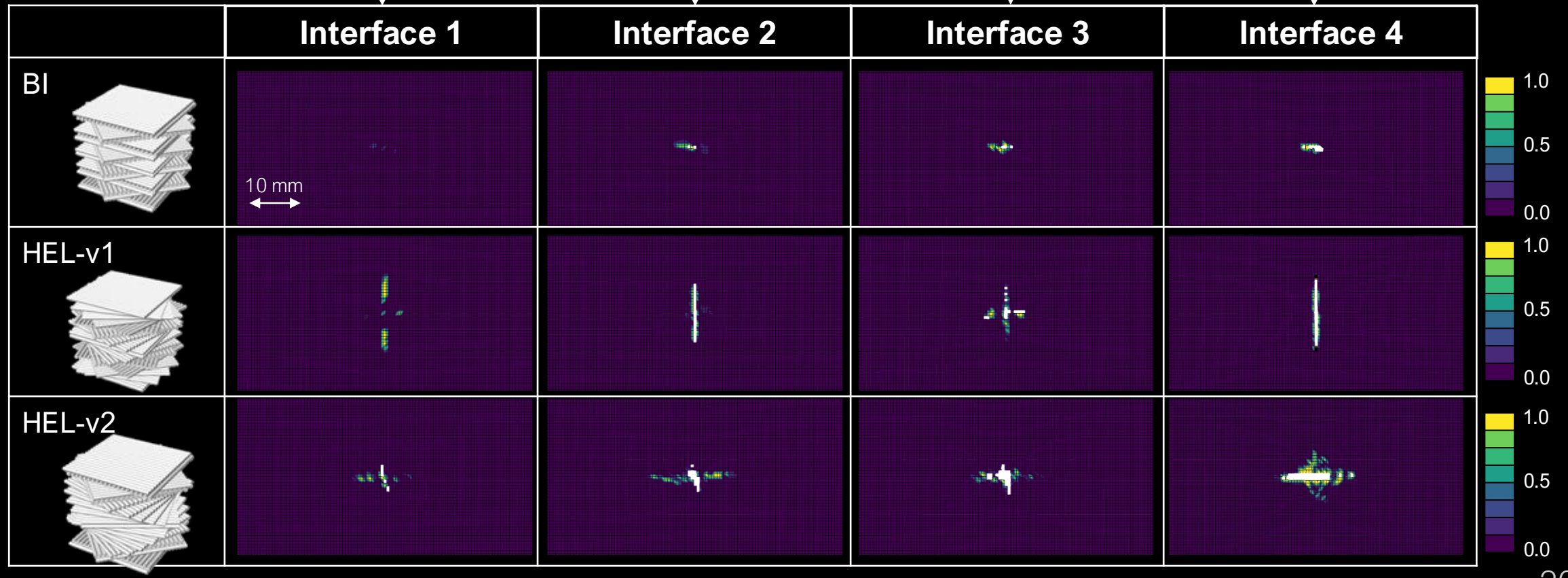
[1] Mencattelli, L. and S.T. Pinho, Herringbone-Bouligand CFRP structures: A new tailorable damage-tolerant solution for damage containment and reduced delaminations. Composites Science and Technology, 2020. 190: p. 108047.

# Results & Discussion

Experimental results | FEA validation | Damage analysis | Thin Wall

- HEL layups show wider inter-ply damage
- Damage spreads across multiple interfaces

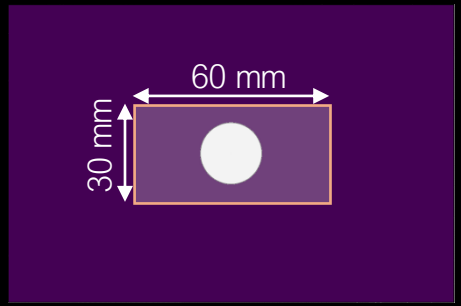


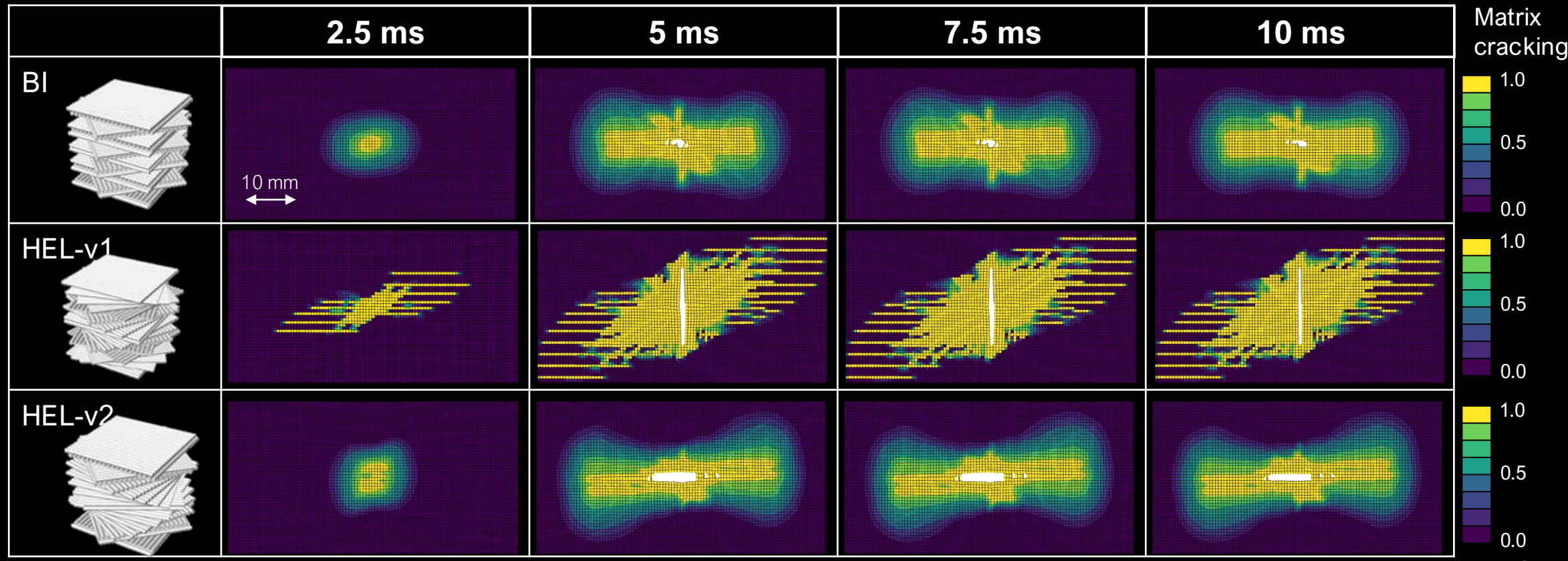



# Results & Discussion

Experimental results | FEA validation | Damage analysis | Thin Wall

- Different layups produce distinct matrix-cracking patterns.
- Both HEL layups spread cracking over a wider area rather than concentrating it.

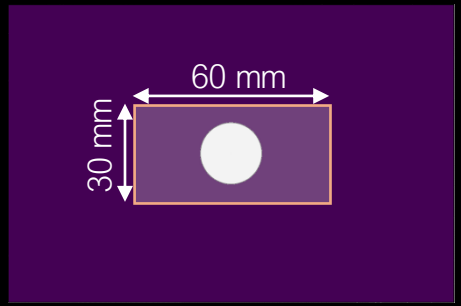


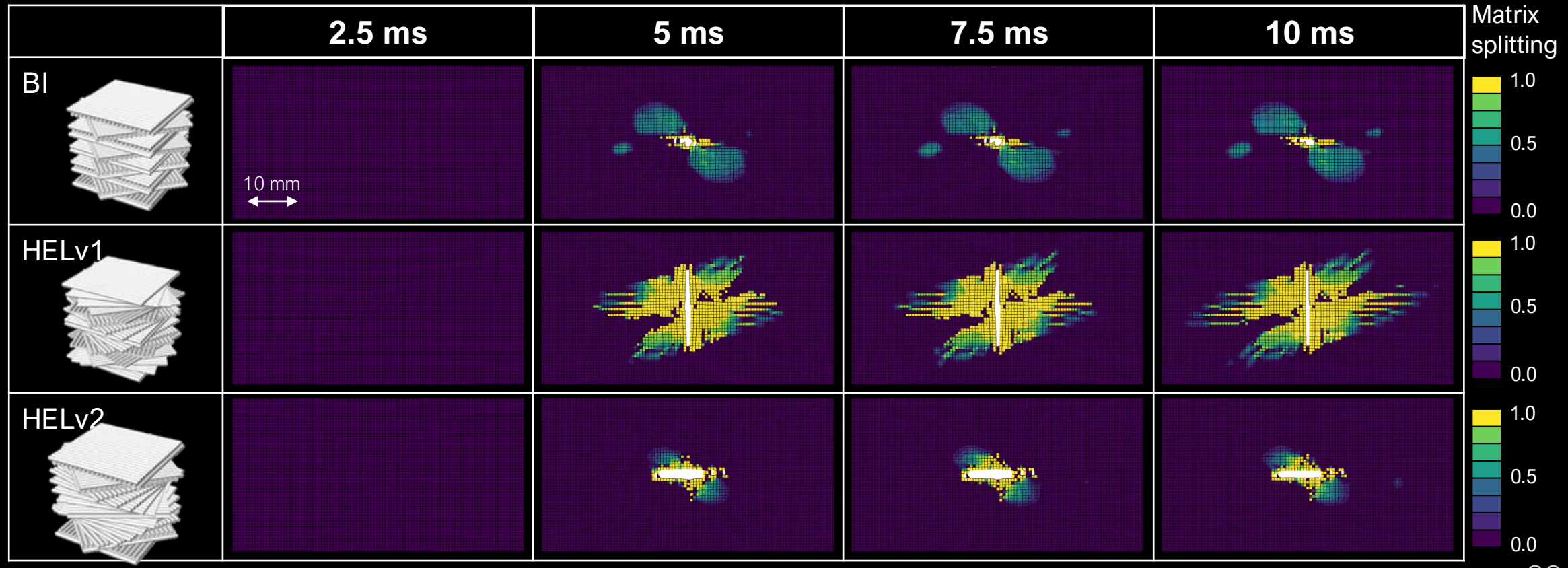



# Results & Discussion

Experimental results | FEA validation | Damage analysis | Thin Wall

- Different layups produce distinct matrix-cracking patterns.
- Both HEL layups spread cracking over a wider area rather than concentrating it.

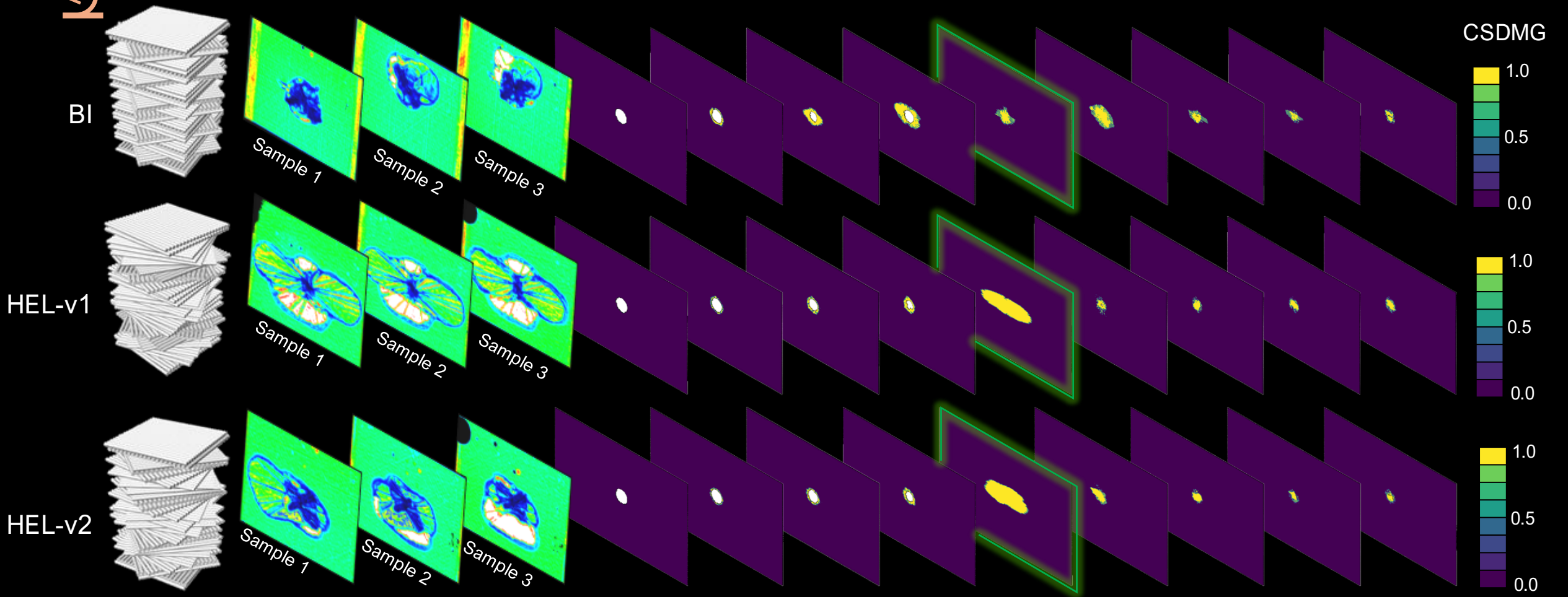


# Results & Discussion

Experimental results | FEA validation | Damage analysis | Thick Wall

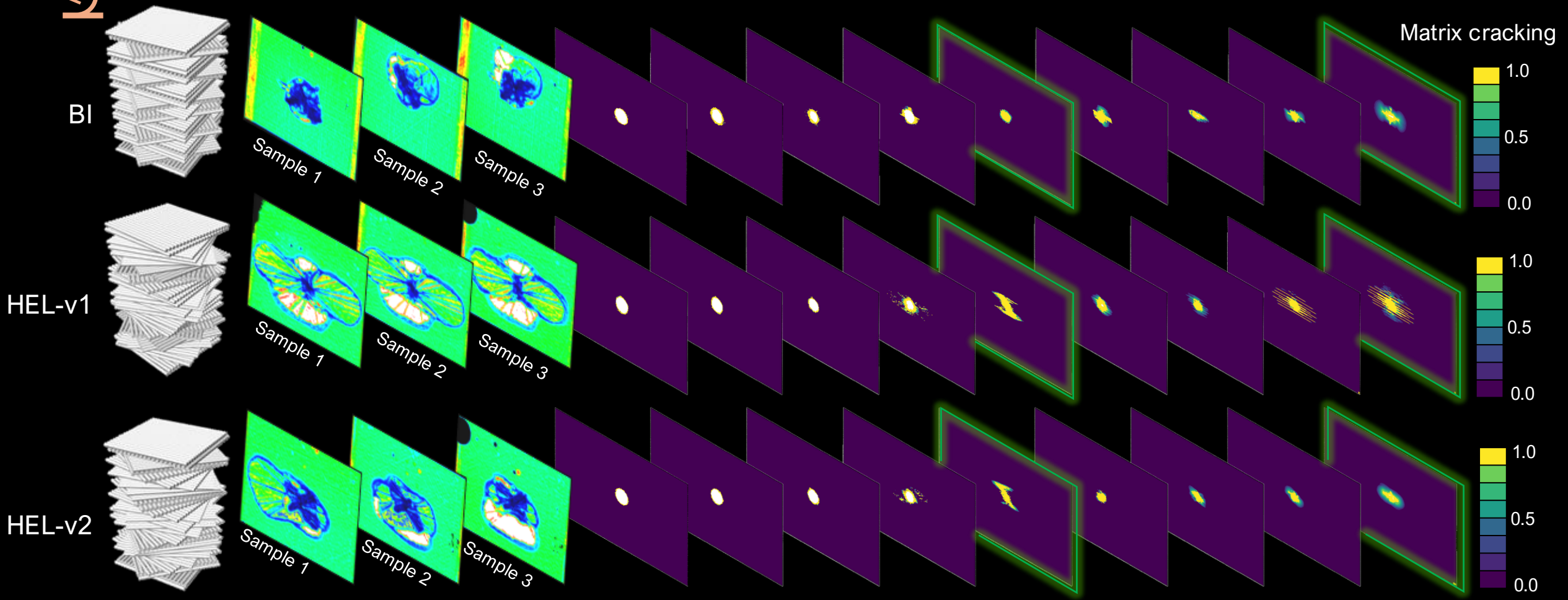
- Compared to BI (thick), both HEL layups show a larger contact-damage region



# Results & Discussion

Experimental results | FEA validation | Damage analysis | Thick Wall

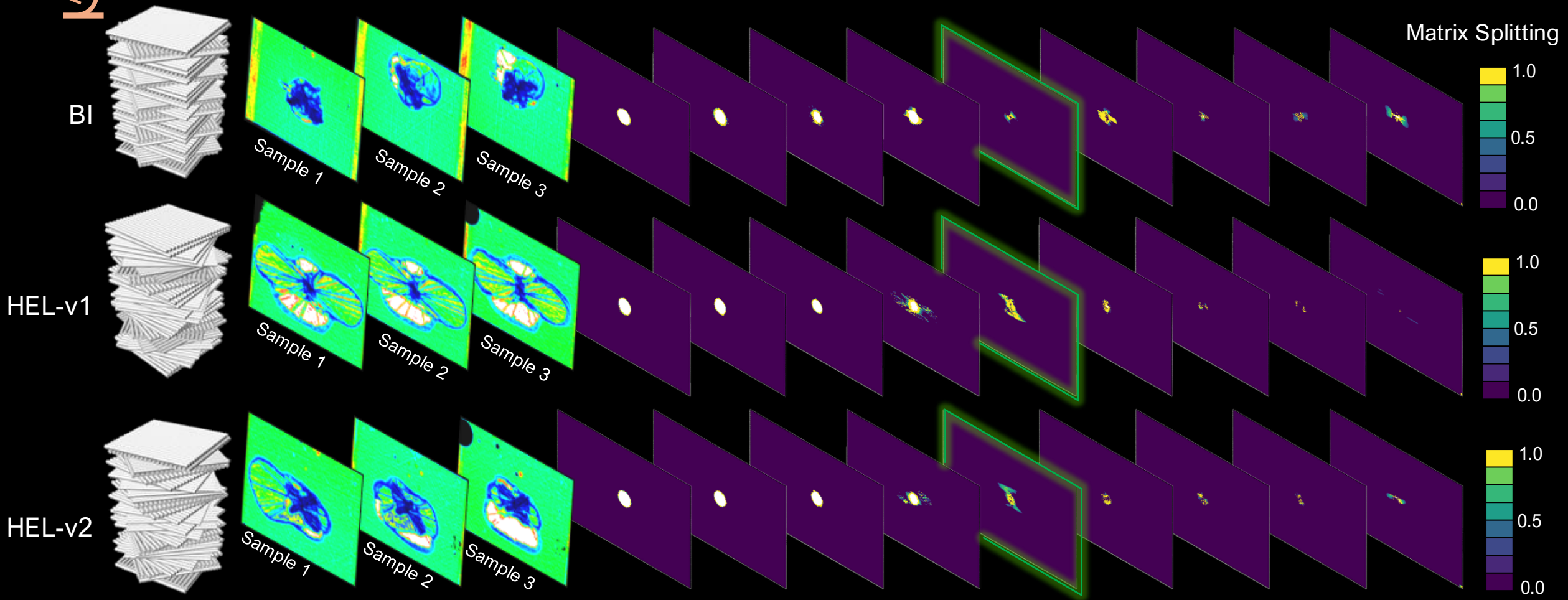
- Compared to BI (thick), both HEL layups show a larger matrix-damage region



# Results & Discussion

Experimental results | FEA validation | Damage analysis | Thick Wall

- Compared to BI (thick), both HEL layups show a larger matrix-damage region



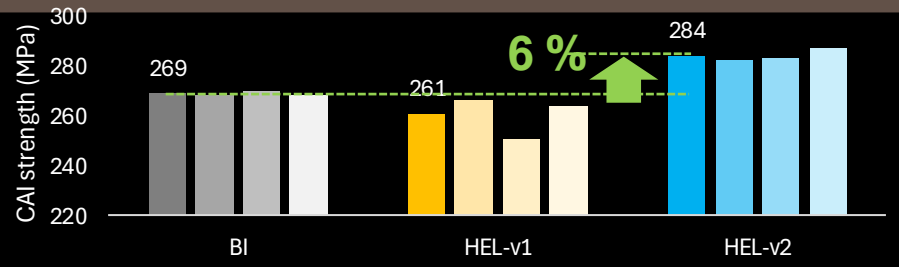
# Results & Discussion

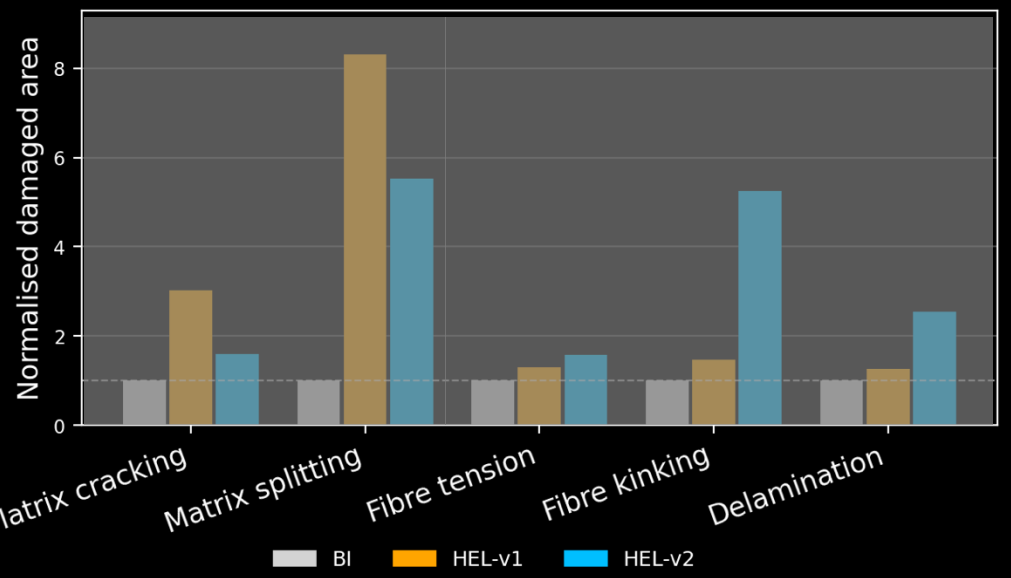
Experimental results | FEA validation | Damage analysis

CAI results

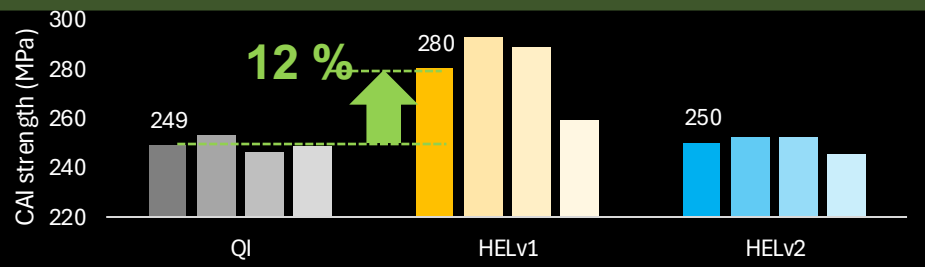
Extracted Damage areas

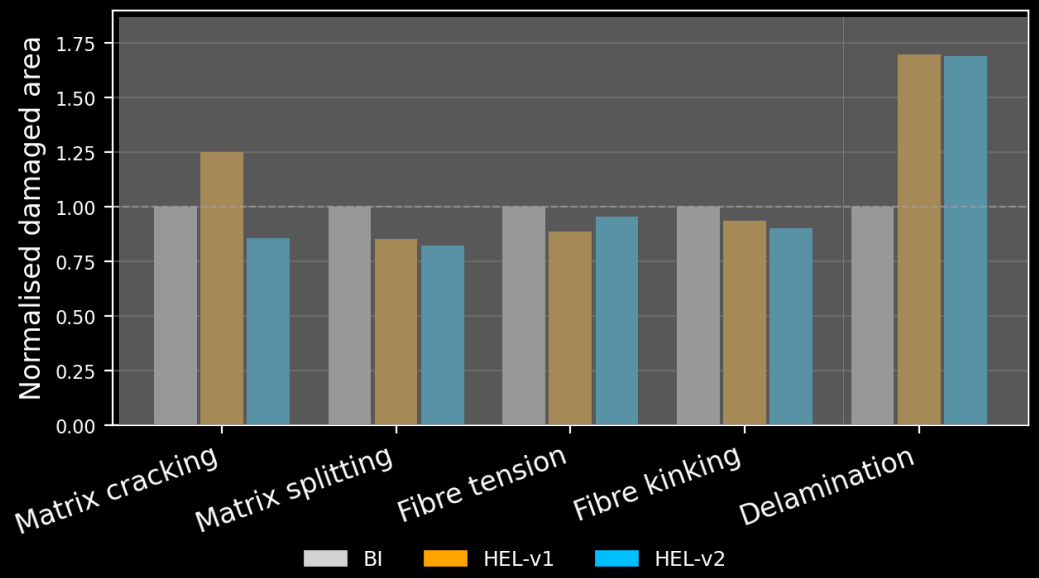
## Thin-wall




## Thick-wall



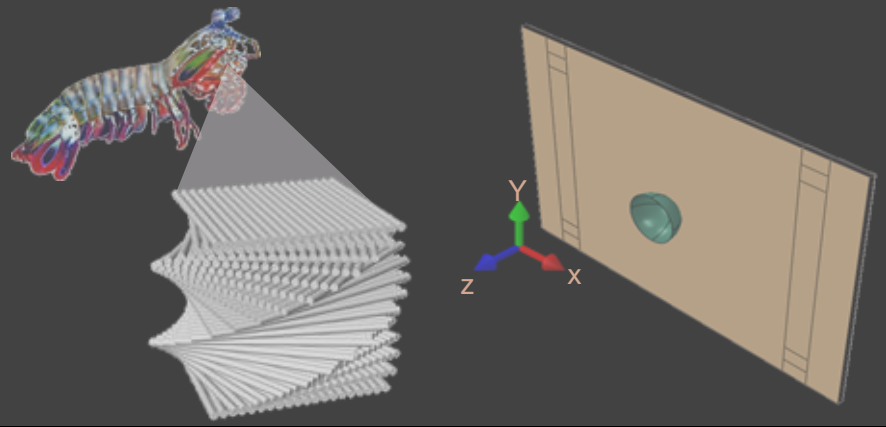

- Distributed delamination absorbs more impact energy;
- The HEL designs can redistribute damage, and thereby improve post-impact damage tolerance.

# Outline

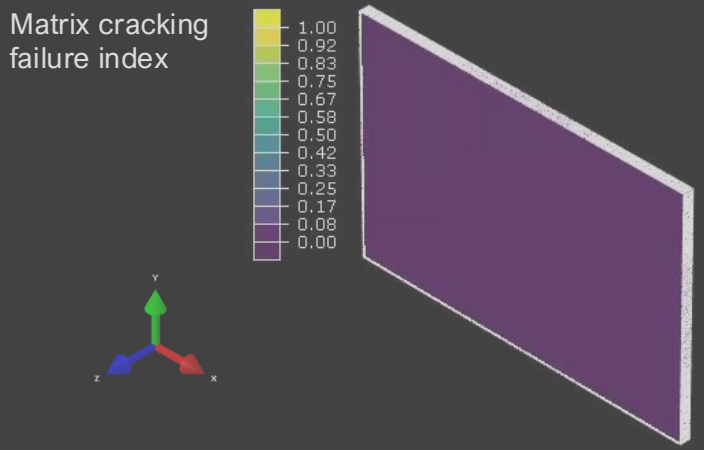
## Background & Motivation



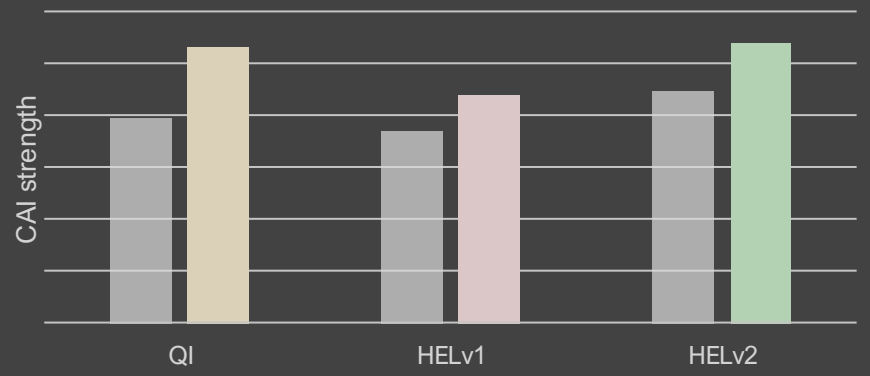

## Design & Methodology




## Results & Discussion




## Summary & Conclusions

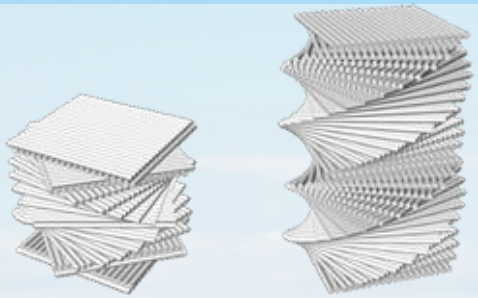


# Finite Element Analysis of Low-Velocity Impact (LVI) Damage in Bio-inspired Helicoid CFRP Laminates for Thin- and Thick- Wall Structures

IMPERIAL

TexComp 9-11 September 2026 Bristol, UK

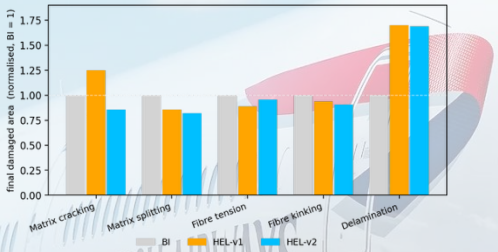
## Summary



Thin wall Thick wall

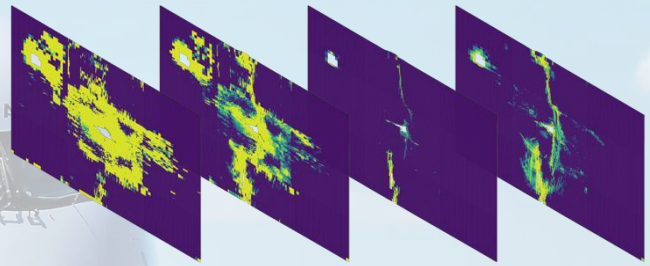
- Designed the BI and helicoidal CFRP layups at two thicknesses;
- Conducted LVI and CAI experiments;
- Developed an FEA framework and validated its predictions aganist the experimental results

## Conclusions




- Helicoidal designs improves CAI strength at both thicknesses (6% for thin-wall, 12 % for thick-wall)
- A smaller pitch angle redistributes impact damage over a larger area, reducing the severity of stress concentrations.
- A more spread delamination after impact can contribute to an improved CAI strength

## Upcoming activities



- Analyse damage propagation in the CAI simulation results to identify the mechanism governing residual strength
- Run FEA simulations using the developed framework for other Helicoidal candidates for various applications


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@hongyan-wang18
https://github.com/Hongyan-Wang
https://pinholab.cc.ic.ac.uk



UKRI-Innovate project - IVI
(Isothermal Variable-volume Infusion)


HELICOID INDUSTRIES LTD

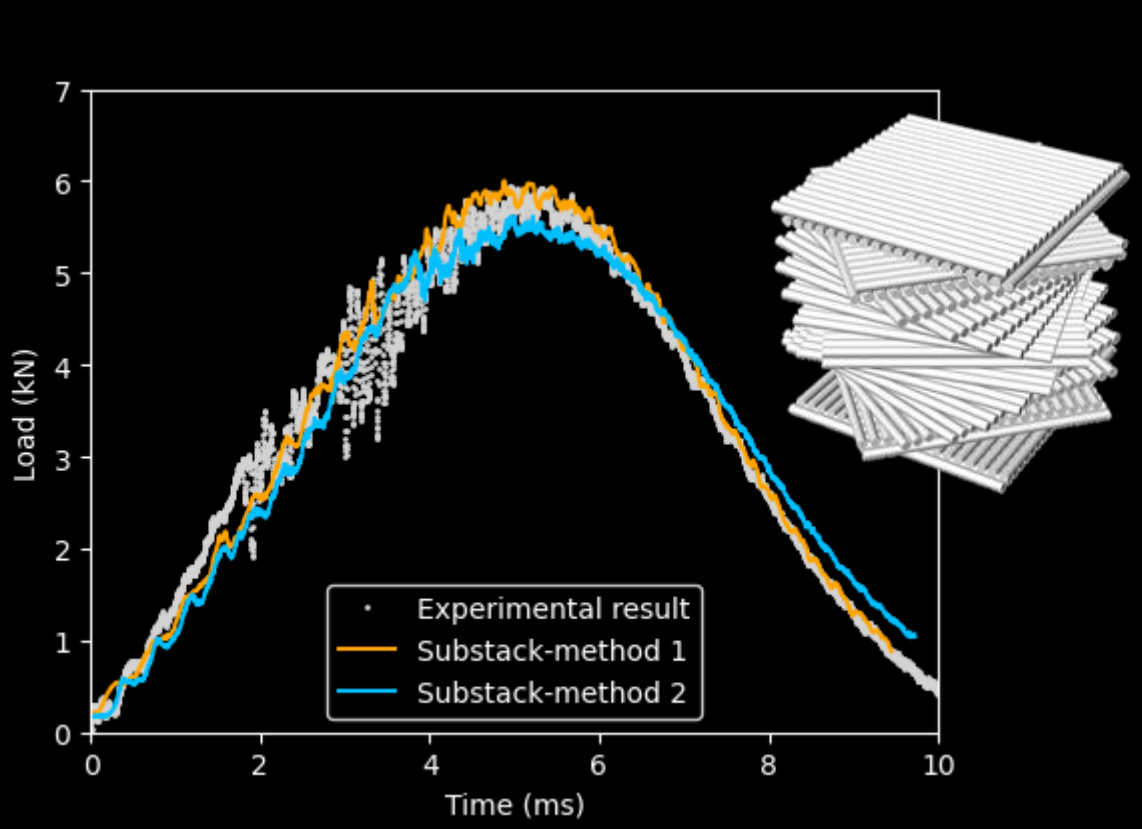
Short Brothers A Boeing Company

Department for Business & Trade

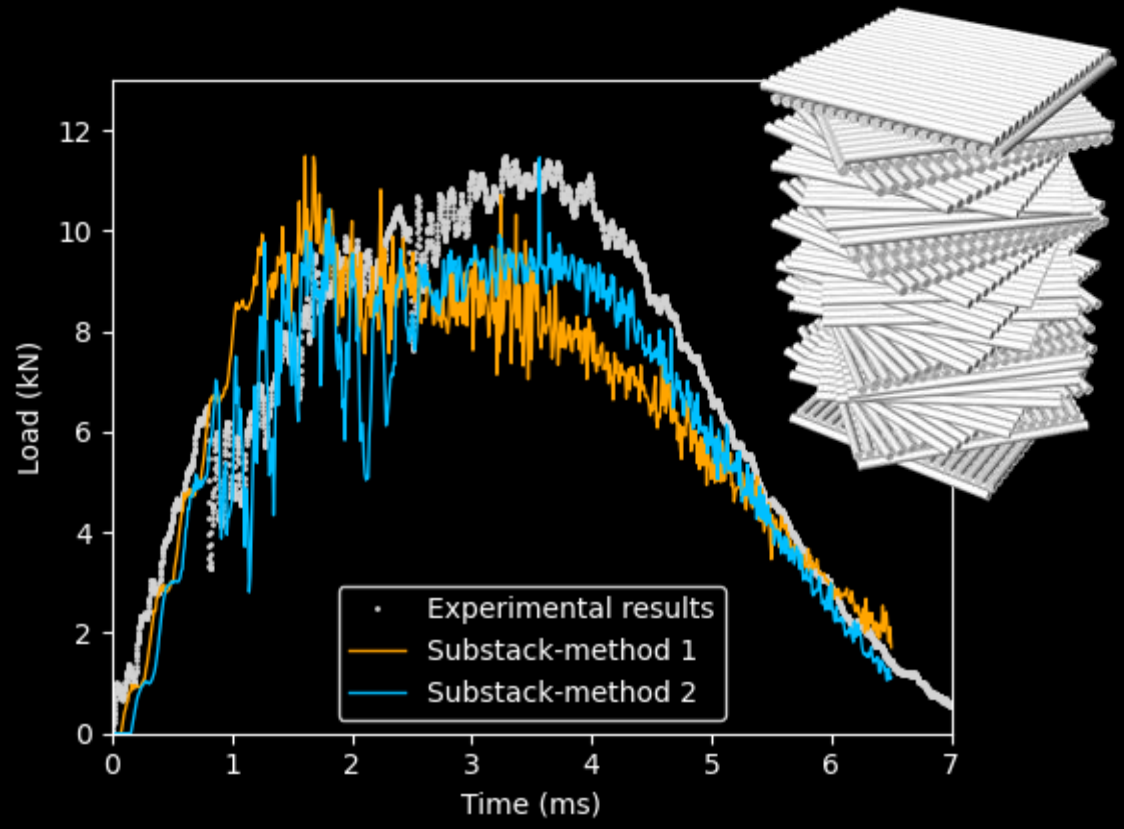
AEROSPACE TECHNOLOGY INSTITUTE

UKRI Innovate UK

# Appendice 1 – substack study




Thin wall (HEL-v2)




Thick Wall (HEL-v2)

- Substack-method 1 comprises of 5 layers per substack
- Substack-method 2 comprises of 3-4 layers per substack

# Appendices 2: LaRc05 model



Matrix cracking:

$$f = \sqrt{\left(\frac{\tau_{\mathrm{T}}}{S_{\mathrm{T}}^{\mathrm{is}} - \eta_{\mathrm{T}}\sigma_{\mathrm{N}}^{*}}\right)^2 + \left(\frac{\tau_{\mathrm{L}}}{S_{\mathrm{L}}^{\mathrm{is}} - \eta_{\mathrm{L}}\sigma_{\mathrm{N}}^{*}}\right)^2 + \left(\frac{\langle\sigma_{\mathrm{N}}\rangle}{Y_{\mathrm{T}}^{\mathrm{is}}}\right)^2}$$

[1]

Matrix splitting:

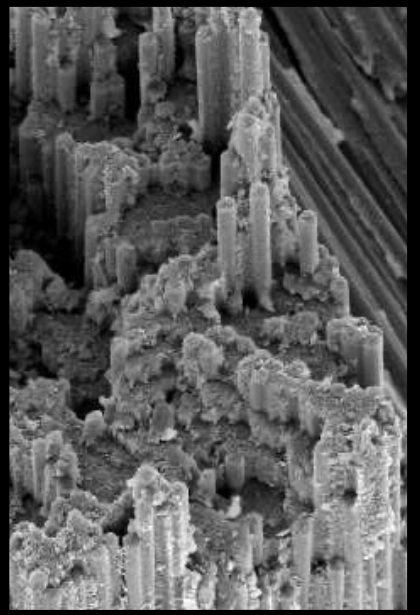
$$f = \sqrt{\left(\frac{\tau_{23}^{\mathrm{m}}}{S_{\mathrm{T}}^{\mathrm{is}} - \eta_{\mathrm{T}}\sigma_{2}^{\mathrm{m}}}\right)^2 + \left(\frac{\tau_{12}^{\mathrm{m}}}{S_{L}^{\mathrm{is}} - \eta_{\mathrm{L}}\sigma_{2}^{\mathrm{m}}}\right)^2 + \left(\frac{\langle\sigma_{2}^{\mathrm{m}}\rangle}{Y_{\mathrm{T}}^{\mathrm{is}}}\right)^2}$$

[1]

Fibre kinking:

$$f = \sqrt{\left(\frac{\tau_{23}^{\mathrm{m}}}{S_{\mathrm{T}}^{\mathrm{is}} - \eta_{\mathrm{T}}\sigma_{2}^{\mathrm{m}}}\right)^2 + \left(\frac{\tau_{12}^{\mathrm{m}}}{S_{\mathrm{L}}^{\mathrm{is}} - \eta_{\mathrm{L}}\sigma_{2}^{\mathrm{m}}}\right)^2 + \left(\frac{\langle\sigma_{2}^{\mathrm{m}}\rangle}{Y_{\mathrm{T}}^{\mathrm{is}}}\right)^2}$$

[1]



Fibre tension:

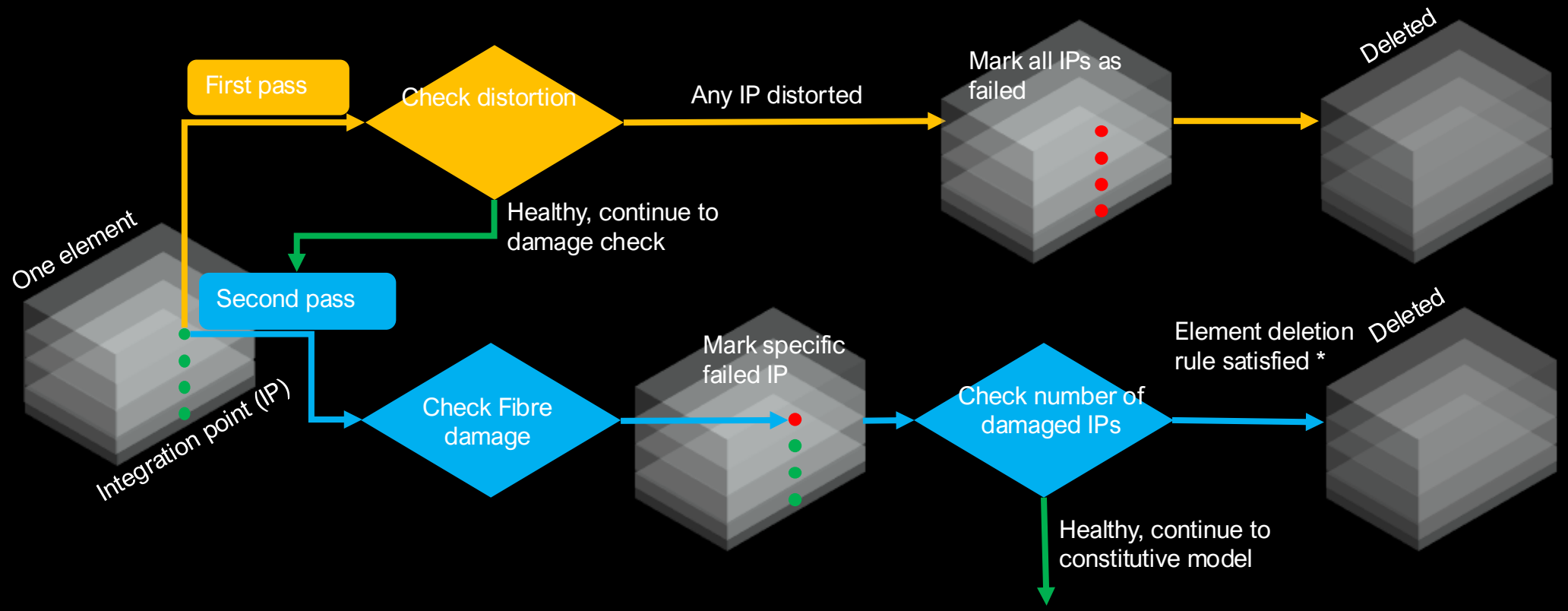
$$f = \frac{\sigma_1}{X^{\mathrm{T}}}$$

[1]

[1] ST Pinho et al. Material and structural response of polymer-matrix fibre-reinforced composites. Journal of Composite Materials. 46, 2313–2341, 2012

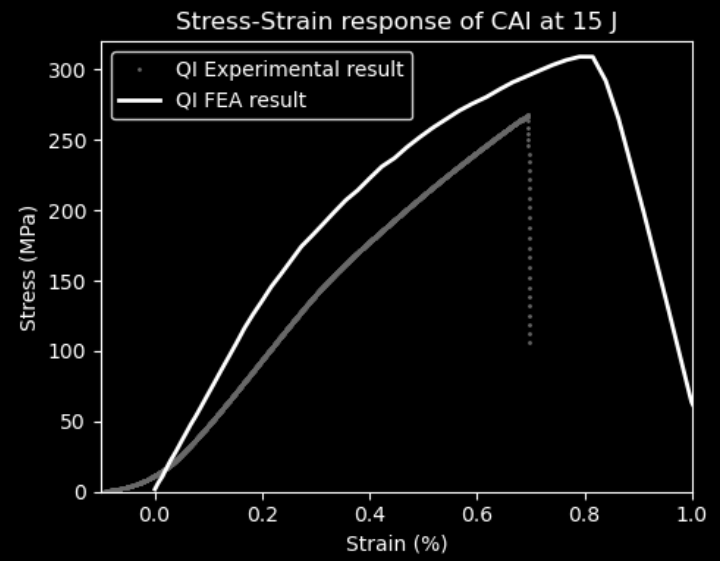
# Appendices 3: How we deal with element distortion

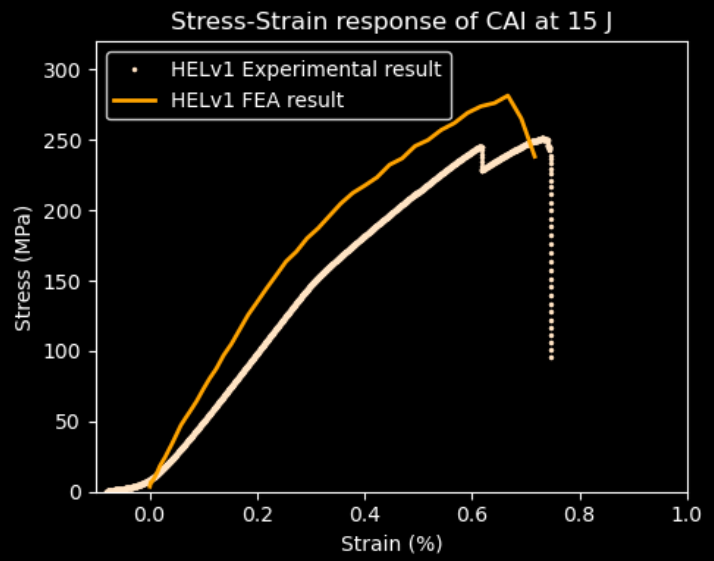
- In default VUMAT subroutines, element deletion is controlled solely by Abaqus, shown schematically as the "Second pass" below.
- We developed a new "First pass" in the VUMAT that checks each element's Jacobian and deletes severely distorted elements before the damage check.

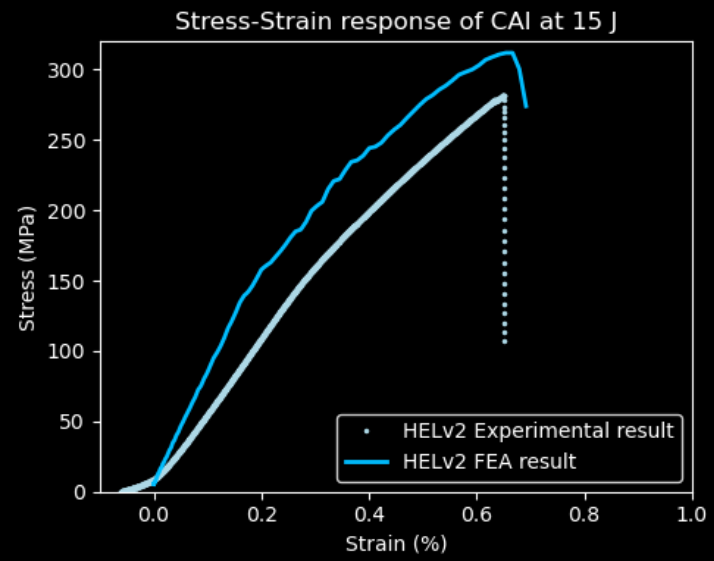


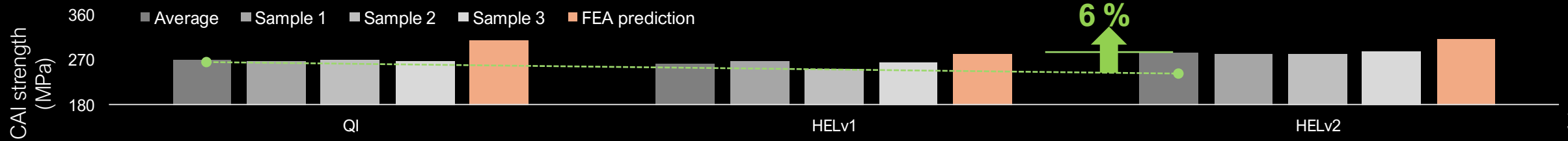
# Appendices 4: CAI results (thin wall)

- FEA prediction of CAI at 15J
  - The CAI stress–strain curves from FEA are compared to the experimental results
  - FEA reproduces the overall CAI response and layup strength ranking
  - Small stiffness differences may reflect test-machine and fixture compliance
  - HELv2 achieves a 6% CAI-strength increase over QI



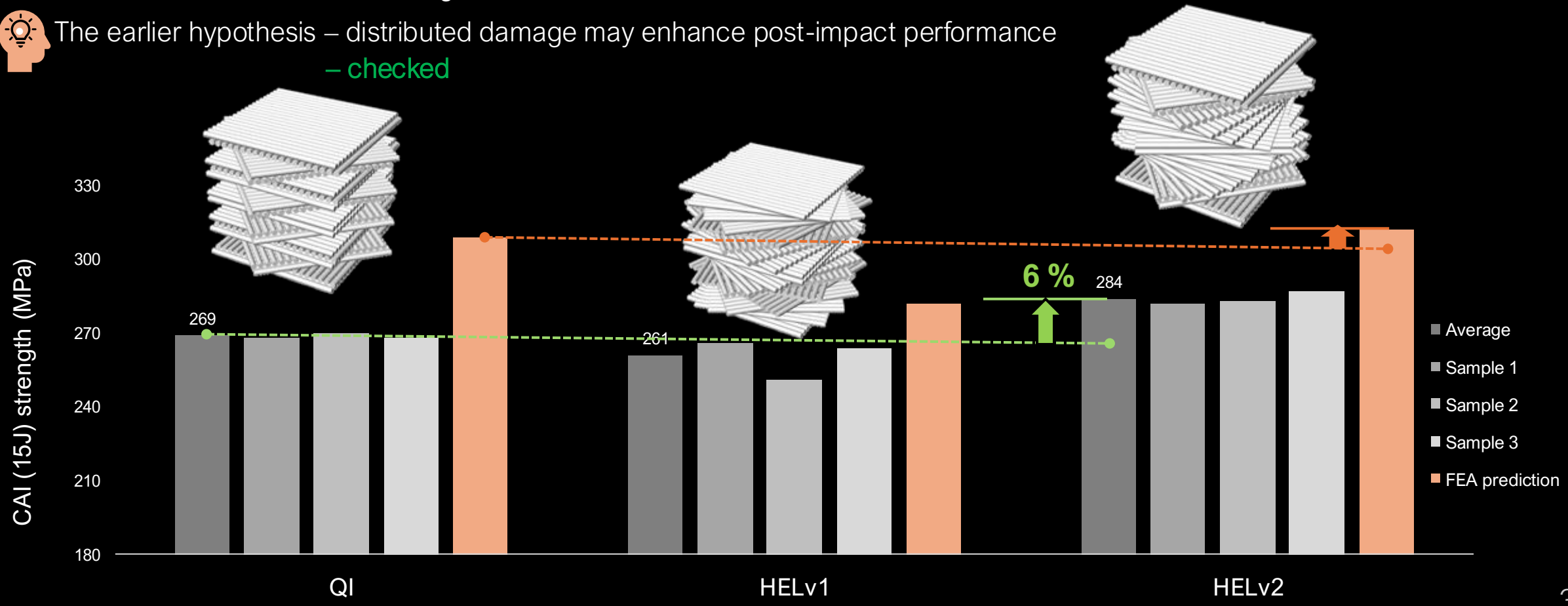



# Appendices 4: CAI results (thin wall)

- FEA prediction of CAI at 15J
  - FEA reproduces the overall CAI response and layup strength ranking
  - Small stiffness differences may reflect test-machine and fixture compliance
  - HELv2 achieves a 6% CAI-strength increase over QI

The earlier hypothesis – distributed damage may enhance post-impact performance – checked



# Appendices 5: substacks division

| BI | HEL-v1 | HELv2 |
|---|---|---|
| 0 | -16 | 0 |
| 90 | -13 | 90 |
| -45 | 6 | 45 |
| 45 | 9 | -45 |
| 0 | 14 | -33 |
| 0 | 31 | -13 |
| 90 | 50 | 7 |
| 0 | 69 | 27 |
| 45 | -69 | 37 |
| -45 | -50 | 42 |
| 0 | -31 | -42 |
| 0 | -14 | -37 |
| 90 | -9 | -27 |
| 0 | -6 | -7 |
| 45 | 13 | 13 |
| -45 | 16 | 33 |
| 0 | 0 | 0 |
| -45 | 16 | 33 |
| 45 | 13 | 13 |
| 0 | -6 | -7 |
| 90 | -9 | -27 |
| 0 | -14 | -37 |
| 0 | -31 | -42 |
| -45 | -50 | 42 |
| 45 | -69 | 37 |
| 0 | 69 | 27 |
| 90 | 50 | 7 |
| 0 | 31 | -13 |
| 0 | 14 | -33 |
| 45 | 9 | -45 |
| -45 | 6 | 45 |
| 90 | -13 | 90 |
| 0 | -16 | 0 |